

THE FLORIDA STATE UNIVERSITY
COLLEGE OF SOCIAL SCIENCES AND PUBLIC POLICY

**TRADING FRONTIER EFFORT FOR COMPETITIVE BALANCE:
KNOWLEDGE SPILLOVERS IN A TWO-STAGE INNOVATION RACE**

By

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Abstract

Rival firms racing to develop the same technology cannot fully contain what they learn. I ask how involuntary knowledge spillovers reshape a winner-take-all innovation race. I build a discrete-time dynamic contest in which two firms race through a Research Stage and a Development Stage, with spillovers active only between firms in the same stage. In the Markov Perfect Equilibrium, these spillovers work in two ways: firms at the frontier free-ride on a rival's effort and invest less, while the value of reaching the frontier rises. Because spillovers are inactive when the race is asymmetric, the laggard responds only to that anticipated value, and so fights harder when there are higher spillovers at the frontier. I prove this transmission when the laggard's research rate does not exceed the leader's development rate, and verify it beyond that region against an independent numerical solution of the model. Spillovers therefore trade frontier effort against competitive balance.

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1 Introduction

Rival firms racing to develop the same technology often cannot fully contain what they learn. Knowledge spills over through channels that sit outside the firms' control. Engineers change employers and carry tacit knowledge with them, especially where non-compete agreements go unenforced (Gilson, 1999; Marx, Strumsky and Fleming, 2009). Patents disclose technical detail as the price of protection. Shipped products can be reverse engineered. Firms clustered in the same region share labor markets, suppliers, and social networks (Saxenian, 1994). These spillovers are set by law, geography, and the nature of the technology, not by the firms themselves. Bloom, Schankerman and Van Reenen (2013) provide reason to take these spillovers seriously: separating the positive technology spillover of a rival's R&D from its negative business-stealing effect, they find that the technology effect is generally stronger (particularly between firms closest in the technology space). The social returns to R&D therefore substantially exceed the private returns.

The Apple Lisa and Macintosh teams of the early 1980s are a concrete example (Isaacson, 2011). The two divisions raced to ship a machine while competing for budget and standing under shared management and infrastructure, with engineers and code moving between them. Progress in one division was imperfectly secret from the other, and the degree of secrecy was set by management, not by the teams. The same structure appears wherever rivals race on proximate technology: what one lab learns, the other partly absorbs, whether either side likes it or not. This example is only a limited illustration: the internal prize was not cleanly winner-take-all, and the spillovers did not always match the model.

Innovation competition is frequently winner-take-all. The prize is a patent, a regulatory approval, or a decisive launch, and the runner-up forfeits its sunk investment. Dynamic contest theory identifies the central danger in such races: the *discouragement effect* (Konrad, 2012). Once behind, the laggard's expected return falls, and it rationally scales back investment. I develop a framework where discouragement reduces effort rather than forcing a full exit. Within the maintained parameter region the laggard never fully quits, but is sys-

tematically out-invested by the leader (Proposition 3), and this asymmetry is what erodes competitive pressure. I ask: *if knowledge spillover between firms close together in the technology space is inevitable, how does it reshape a winner-take-all race — deepen free-riding, sustain competition, or both?*

To answer this question, I develop a discrete-time dynamic contest between two ex-ante symmetric firms. Winning requires two sequential breakthroughs: one to complete a Research phase and one to complete a Development phase. The first firm to complete Development wins the prize W ; if both firms complete it simultaneously, a fair coin flip allocates the prize, so each expects $W/2$. The central feature of this model is a state-dependent spillover parameter $\beta_\omega \in [0, 1]$. Each firm’s effective effort is $x_i + \beta_\omega x_j$, and the spillover is active only when both firms occupy the same stage.

I solve the game for its Markov Perfect Equilibrium by backward induction. It shows how firms adjust investment to their environment, and — most importantly — how a spillover in one stage of the race alters competitive incentives in the others.

Higher development-stage spillovers depress frontier effort but raise the value of reaching the frontier. Higher β_{11} lowers frontier effort: when both firms are in the tied development state, each free-rides on what spills over from the other. At the same time, higher β_{11} raises the value of the tied state itself, V_{11} , because a firm at the frontier shares in what its rival produces there. This second channel is anticipatory rather than direct. Spillovers are inactive in the asymmetric states, so the laggard receives no benefit from the leader’s effort; it fights harder only because the tied state it is racing toward is more valuable. I prove that when the laggard’s research rate does not exceed the leader’s development rate ($\mu \leq \lambda$), the laggard’s equilibrium effort is strictly increasing in the development-stage spillover (Corollary 1). The proof needs only $x^* < \sqrt{2} z^*$, which holds at every equilibrium I scan, including all 13,176 draws where the leader’s effort advantage fails, so the result is a theorem well past the region where I can prove its hypothesis (Section 5).¹

¹Replication code, result logs, and figures: https://github.com/iellahi/Honors_Thesis

The model applies to several real-world settings. A primary application is the internal R&D tournament: two divisions racing for budget, mandate, or promotion, where shared management, shared infrastructure, and internal mobility make progress imperfectly secret. Since β here is set by the management, this setting is particularly useful in analyzing optimal knowledge spillover. The race between the Apple Lisa and Macintosh teams serves as a clear historical example of this dynamic.

Common ownership amplifies these spillovers. Shared institutional owners are associated with cross-citation and collaboration among portfolio firms (Kostovetsky and Manconi, 2020), and theory predicts that innovation rises with common ownership precisely when technology spillovers dominate product market rivalry (López and Vives, 2019; Antón et al., 2024). Because the empirical literature is debated, this thesis does not use common ownership as its core foundation. Instead, it is a modern mechanism that makes spillovers more likely.

The analysis is organized around the equilibrium and a series of comparative-statics. For each state of the race, I ask how equilibrium efforts and values move with prize W , the cost c , the breakthrough rates μ and λ , and the spillovers β_{00} and β_{11} . I prove most results as propositions and support others numerically. Table 1 in Section 3 summarizes the main comparative statics in one place. The broad pattern is intuitive: a spillover encourages free-riding and depresses effort in the stage where it occurs $(1, 1)$, but makes reaching that stage more valuable — so firms invest more aggressively in earlier stages to catch up.

The rest of the thesis is structured as follows. Section 2 reviews the related literature. Section 3 sets up the model. Section 4 characterizes the Markov Perfect Equilibrium and proves the analytical comparative statics for each state. Section 5 runs an independent numerical consistency check on those proven closed forms—guarding against algebra error rather than providing evidence—and then examines the comparative statics that resist analytical proof. Section 6 concludes and discusses limitations.

2 Literature Review

This section covers the intersection of the two literatures in this thesis — empirical work on R&D knowledge spillovers and the theory of dynamic contests. Internal tournaments serve as a supporting case for the model, and common ownership acts as an amplifier.

2.1 R&D Spillovers and Channels

This thesis builds on Bloom, Schankerman and Van Reenen (2013), who separate two distinct effects of a rival’s R&D: a positive technology spillover and a negative business-stealing effect. They find that the technology spillover dominates, so the social returns to R&D exceed the private returns, and that spillovers operate most strongly between firms that are close in technology space. This maps directly onto the model. The same-stage spillover parameter β_ω captures spillovers between technologically close rivals, and the winner-take-all prize captures business stealing.

Two older papers micro-found this same-stage restriction. Jaffe (1986) shows that spillovers concentrate among technological neighbors. Cohen and Levinthal (1990) argue that a firm can only absorb external knowledge if it has the internal capability to understand it, which they call *absorptive capacity*. My model implements this with the 0/1 same-stage condition—a firm absorbs its rival’s effort only after reaching the same technological stage.

Several institutional channels make these spillovers involuntary from the firms’ perspective. Engineers move between employers where non-compete agreements go unenforced (Gilson, 1999; Marx, Strumsky and Fleming, 2009). Firms in the same region share labor markets, suppliers, and social networks (Saxenian, 1994). Patents disclose technical detail as the price of protection, and shipped products can be reverse engineered. One caveat: I treat the Silicon Valley versus Route 128 comparison as a mechanism, not a causal fact.

Finally, treating β as exogenous has theoretical support. d’Aspremont and Jacquemin (1988) model R&D competition with an exogenous spillover parameter, and López and Vives

(2019) is a modern version of that framework.²

2.2 Dynamic Contests and Discouragement Effect

The theoretical basis of this thesis is the literature on dynamic contests. Taylor (1995) establishes the foundational model of research tournaments, and Baye and Hoppe (2003) generalizes the underlying contest structures.

A central tension in this literature is the discouragement effect (Konrad, 2012): once behind, a laggard’s expected return falls and it scales back investment. Möller (2012) and Beviá and Corchón (2013) further explore how early asymmetries damage competitive balance in contests.

The closest theoretical work is step-by-step innovation. Aghion et al. (2001) distinguish leveled states, where firms are neck and neck, from unleveled states with a leader and a laggard, and study the incentive to escape competition. Aghion et al. (2005) build on this framework and document an inverted-U relationship between competition and innovation — more competition boosts innovation when competition is low but stifles it when competition is already high. My model connects to this directly: the laggard’s catch-up incentive works through the value of reaching the tied state, V_{11} , which is the same idea as the value of reaching the leveled state.

Further closely related works on contest design and disclosure include Aoyagi (2010), Bimpikis, Ehsani and Mostagir (2019), Che and Gale (2003), Ales, Cho and Korpeoglu (2017), Sela (2011), and Letina (2016).

2.3 Internal R&D Tournaments

Internal tournaments — divisions inside the same firm competing for an internal prize — are the supporting case for the model. Lazear and Rosen (1981) establish the foundational

²Amir (2000) identifies a well-definedness issue in the original d’Aspremont–Jacquemin specification of spillovers; see the discussion there.

theory of rank-order tournaments, where relative performance determines the prize. Jia, Tian and Zhang (2016) provide empirical evidence that tournament incentives raise firm innovation.

Nelson (1961) explores parallel R&D as a hedge against technological uncertainty. Boudreau, Lacetera and Lakhani (2011) looks at how contest structure and additional competitors impact effort.

2.4 Common Ownership

Common ownership enters this thesis as an amplifier of spillovers, not as a foundation for the model. On diffusion, Kostovetsky and Manconi (2020) find that shared institutional ownership is associated with cross-citation of patents and collaborative research among portfolio firms. On theory, López and Vives (2019) show that overlapping ownership leads firms to internalize technology spillovers. Antón et al. (2024) establish the sign structure: innovation rises with common ownership precisely when technology spillovers dominate product market rivalry. Related mechanisms appear in Aghion, Van Reenen and Zingales (2013), He and Huang (2017), and Li, Liu and Mai (2021).

The broader common ownership literature is contested. Azar, Schmalz and Tecu (2018), Ederer and Pellegrino (2023), and Antón et al. (2022) document anti-competitive effects of common ownership, while O’Brien and Waehrer (2017) and Dennis, Gerardi and Schenone (2022) challenge the empirical methods behind some of the headline results. Because the evidence remains disputed, I use common ownership only as a modern illustration that can increase the level of knowledge spillovers between firms.

2.5 Contribution

Building on the empirical literature on R&D spillovers, I develop a dynamic multi-stage model of how involuntary knowledge spillover reshapes behavior in an innovation race, with the same-stage restriction grounded in absorptive capacity. Relative to the static tradition

from d’Aspremont and Jacquemin (1988), the spillover here is state-dependent and its effects are dynamic: β enters one state directly but travels to every other state through continuation values. The multi-stage structure is what creates the question, since a spillover at the frontier changes behavior far from the frontier.

On the theory side, I characterize how state-dependent spillovers act on the discouragement effect: not by rescuing the laggard directly, but through the anticipated value of the tied state. This channel is a theorem, not a numerical pattern. The tied-state value rises with the development-stage spillover (Proposition 2), the laggard’s equilibrium effort rises with the tied-state value (Proposition 5), and the chain closes whenever the laggard’s research rate does not exceed the leader’s development rate (Corollary 1). In the language of the step-by-step innovation literature (Aghion et al., 2001), V_{11} is the leveled-state value and the result shows that the laggard’s incentive to exert effort is monotone in the level of knowledge spillovers in the symmetric state.

Where the spillover level is a designer’s choice—as it is inside a firm—Proposition 1 and Corollary 1 sign both sides of the trade-off it creates: less effort at the frontier, and a laggard that stays in the race.

3 Model

There are two ex-ante symmetric firms, $i \in \{A, B\}$, racing in discrete time to complete a two-stage innovation: a Research breakthrough followed by a Development breakthrough. The first firm to complete Development wins a prize $W > 0$; the loser forfeits all sunk costs. The state of the race is the pair of stage positions $\omega \in \{(0, 0), (1, 0), (0, 1), (1, 1)\}$, where $\omega_i = 0$ (1) places firm i in the Research (Development) stage. Firms are risk neutral and do not discount ($\delta = 1$), which removes any discouragement arising purely from prize depreciation.

Each period, every firm succeeds or fails independently: conditional on efforts, break-

through realizations are independent Bernoulli draws, so both firms succeed with probability $P_i P_j$ and neither with probability $(1 - P_i)(1 - P_j)$. Allowing both firms to succeed in the same period is the natural specification for two labs drawing independently; the alternative—forcing at most one success per period—would need an ad hoc rule for who succeeds when both draw through. The only tie this creates is the terminal one in $(1, 1)$, resolved by a coin flip that is economically neutral under risk neutrality. This also opens up transitions that a mutually-exclusive-breakthrough model cannot generate. The state moves as follows. In $(0, 0)$, if only one firm achieves a research breakthrough the state moves to $(1, 0)$ or $(0, 1)$; if both succeed simultaneously (probability $P_i P_j$) it jumps directly to $(1, 1)$; if neither succeeds it repeats. In $(1, 0)$ —where I call the firm in Development the *leader* and its rival the *laggard*—a development breakthrough by the leader ends the game and takes precedence over any simultaneous research breakthrough by the laggard (total probability P_L); a laggard-only breakthrough moves the state to $(1, 1)$; otherwise it repeats. In $(1, 1)$, if exactly one firm achieves a development breakthrough it wins W ; if both do, a fair coin flip allocates the prize, equivalent under risk neutrality to sharing it in expectation ($W/2$); otherwise the state repeats. Figure 1 summarizes the transitions.

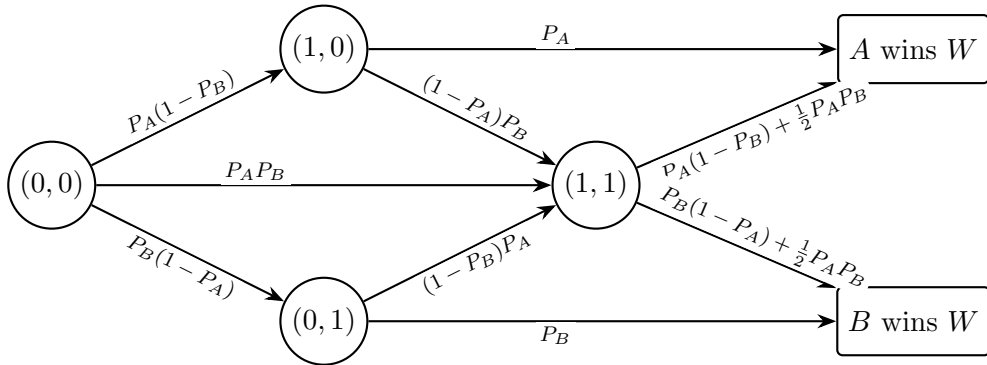


Figure 1: State transitions of the innovation race. P_i denotes firm i 's per-period breakthrough probability in the current state; with probability $(1 - P_A)(1 - P_B)$ neither firm succeeds and the state persists. Simultaneous research breakthroughs move the game directly from $(0, 0)$ to $(1, 1)$. In the asymmetric states the leader's development breakthrough takes precedence, so the leader wins with total probability P_A (resp. P_B) regardless of the laggard's simultaneous outcome. A simultaneous development breakthrough in $(1, 1)$ is resolved by a fair coin flip.

Knowledge spills over between the firms, and how much depends on where they stand.

Spillovers are exogenous, state-dependent, and active only when both firms occupy the same stage: $\beta_\omega \in [0, 1]$, with β_{00} in the research state, β_{11} in the development state, and $\beta = 0$ in the asymmetric states. Each firm exerts effort x_i , and effective effort when both firms share a stage is

$$e_i(\omega) = x_i + \beta_\omega x_j.$$

The same-stage restriction is a simple, all-or-nothing version of absorptive capacity (Cohen and Levinthal, 1990): absorbing a rival's advances requires having reached the relevant stage, so capability enters as a 0/1 condition rather than a smooth scale, with β_{00} and β_{11} free to differ. Empirically, spillovers concentrate among technological neighbors (Jaffe, 1986; Bloom, Schankerman and Van Reenen, 2013). The intuition is direct: a firm stuck in basic research cannot internalize its rival's late-stage advances, while two firms working through the same stage can hardly avoid it. Within a stage, additive absorption is a linear approximation, which I return to in the limitations (Section 6). The zero-spillover benchmark ($\beta_{00} = \beta_{11} = 0$) is a special case.

Effort maps to success probability through a linear contest success function, $P_i(\omega) = \gamma_i e_i(\omega)$, where $\gamma_i = \mu$ when firm i is in Research and $\gamma_i = \lambda$ when it is in Development. Effort costs are quadratic and common across stages, $C(x_i) = \frac{c}{2} x_i^2$ with $c > 0$; this convexity is what makes effort interior.

Three restrictions are maintained throughout.

Assumption 1 (A1). $x_i \in [0, 1]$ and $\lambda, \mu < \frac{1}{2}$, so that $P_i = \gamma_i(x_i + \beta_\omega x_j) < 1$ even as $\beta_\omega \rightarrow 1$.

Assumption 2 (A2). $\lambda W \leq c$, which guarantees an interior, valid-probability equilibrium in the development state.

Assumption 3 (A2'). $\mu W \leq c$, the research-state analogue of Assumption 2; I refer to it as (A2'), and to Assumptions 1 and 2 as (A1) and (A2).

Assumption (A2) has a simple economic reading: it bounds the per-period expected prize relative to marginal cost, and so restricts attention to races where effort is an *interior* choice. Outside (A2) the model remains well-defined—the equilibrium is the corner $x^* = 1$, both firms at maximal effort—but strategically uninteresting, since effort is pinned at the maximum and no margin is left for spillovers to move. (A2) is also conservative: numerically, the true interior/corner frontier lies at $\lambda W/c \approx 1.5\text{--}1.7$, so interior equilibria persist in a band beyond the assumption (Section 5). The propositions below cover the interior regime, where the central trade-off lives.

Within each period, firms choose efforts simultaneously, the independent Bernoulli realizations occur, and the state updates. The generic Bellman equation is

$$V_i(\omega) = -\frac{c}{2}x_i^2 + P_i(1 - P_j)[\cdot] + P_j(1 - P_i)[\cdot] + P_iP_j[\cdot] + (1 - P_i)(1 - P_j)V_i(\omega),$$

where the bracketed terms are the continuation values of the corresponding transitions. I solve for the Markov Perfect Equilibrium by backward induction, restricting attention to equilibria that are symmetric in the symmetric states $(0, 0)$ and $(1, 1)$ as a form of selection.³ Within this class the equilibrium is unique; Section 4 and Appendix C establish this state by state.

Comparative Statics. Section 4 solves the game state by state and organizes the analysis around a fixed set of comparative statics. Because the tied-state value V_{11} itself moves with the parameters, each comparative static in the asymmetric states has a *direct* channel (holding V_{11} fixed) and a *total* channel (letting V_{11} adjust), and the two can in principle disagree. Every claim carries an explicit epistemic status: proven, or numerical. Table 1 previews the full program. The asymmetric-state equilibrium has no closed form, but it is clear from its best-response curves (Figure 3 in Section 4), and its comparative statics are exact where marked.

³A full examination of asymmetric equilibria is beyond the scope of this thesis.

State	Object	Parameter	Sign	Status	Channel
(1, 1)	x_{11}^*	W, λ	+	proven (Prop. 1)	direct
(1, 1)	x_{11}^*	c, β_{11}	−	proven (Prop. 1)	direct
(1, 1)	V_{11}	β_{11}, W, c	+	proven (Prop. 2)	direct
(1, 1)	V_{11}	λ	−	proven (Prop. 2)	direct
(1, 0)	x^* (laggard)	β_{11}	+	proven, $\mu \leq \lambda$ (Cor. 1)	via V_{11} only
(1, 0)	z^* (leader)	β_{11}	ambiguous ($\approx 96\%$ −)	sign exact (Lem. 8)	via V_{11} only
(1, 0)	z^*, x^*	c	−	numerical (uniform)	total = direct
(1, 0)	z^*	λ	+	numerical (99.99%)	total = direct
(1, 0)	x^*	λ	+, rare reversals ($\approx 2\%$)	numerical	total; indirect
(0, 0)	x_{00}^*	V_{10}, V_{11}	+	proven (Prop. 4)	V_{11} : direct-jump
(0, 0)	x_{00}^*	V_{01}	−	proven (Prop. 4)	direct

Table 1: The comparative-statics program of Sections 4–5. “Direct” holds continuation values fixed; “total” lets V_{11} adjust. Statuses reference Section 5 and Appendix C.

4 Equilibrium Analysis

Every channel in the model goes through the tied-state value V_{11} , so I solve the state that determines it first and work backward. Accordingly, the analysis proceeds via backward induction, beginning with the symmetric development stage, (1, 1). The derivations follow a consistent set of steps. In each case, I specify the Bellman equation and use it to derive the first-order condition (FOC). In the symmetric states, imposing symmetry on the equilibrium effort yields a quadratic that characterizes the solution. In the asymmetric states, the FOCs and the Bellman equations yield two quadratic equations that characterize the best-response functions for the leader and the laggard. These are solved numerically for a range of parameter specifications in Section 5. Although there is no closed-form, analytical solution for the equilibrium efforts, a range of comparative statics follows from the first-order conditions.

4.1 Symmetric Development State (1, 1)

Both firms are in the development phase, so $P_i = \lambda(x_i + \beta_{11}x_j)$. The Bellman equation for firm i is:

$$V_{11} = -\frac{c}{2}x_i^2 + P_i(1 - P_j)W + P_iP_j\frac{W}{2} + (1 - P_i)(1 - P_j)V_{11}. \quad (1)$$

Throughout this subsection, let $M \equiv \lambda(1 + \beta_{11})$, so that under symmetric play $P = Mx$. Maximizing with respect to x_i and imposing symmetry ($x_i = x_j = x$) yields the first-order condition:

$$cx = \lambda W - MV_{11} - M^2x \left(\frac{W}{2} - V_{11} \right). \quad (2)$$

Equating the FOC-implied value with the Bellman-implied value (Appendix A) reduces the equilibrium to a single quadratic in x :

$$c\lambda(1 + \beta_{11})x^2 - (3c + \lambda^2W(1 - \beta_{11}^2))x + 2\lambda W(1 - \beta_{11}) = 0. \quad (3)$$

Let $u \equiv \lambda^2W(1 - \beta_{11}^2)$ denote the prize term in the middle coefficient.

Lemma 1 (Discriminant). *The discriminant of (3) satisfies $\Delta = 9c^2 - 2cu + u^2 = (u - c)^2 + 8c^2 > 0$ unconditionally.*

Both roots are therefore real for all admissible parameters.

Lemma 2 (Root selection). *Under (A1)–(A2), the two roots of (3) straddle $1/M$. The additive root implies a breakthrough probability $P > 1$ and is infeasible. The unique valid equilibrium effort is the subtracted root,*

$$x_{11}^* = \frac{3c + u - \sqrt{\Delta}}{2c\lambda(1 + \beta_{11})} \in (0, 1). \quad (4)$$

The additive root is infeasible iff $u < 2c$, which (A2) guarantees. Absent (A2), a corner regime exists (for example, $\lambda = 0.49$, $c = 0.1$, $W = 50$ gives a subtracted root of approximately 4.05, so the equilibrium is the corner $x^* = 1$). Full details are in Appendix A.

At the boundary $\beta_{11} = 1$, the constant term of (3) vanishes, so $x_{11}^* = 0$ exactly: full knowledge diffusion leads to full free-riding.⁴ Substituting the FOC back into the Bellman

⁴The boundary is a limit point, not a well-posed case: at $x = 0$ the Bellman equation degenerates to $V_{11} = V_{11}$ and every value solves it. The reported $W/2$ is what the first-order condition pins down, and the limit of (5). Results below assume $\beta_{11} < 1$.

equation gives the value function:

$$V_{11} = \frac{cx_{11}^*}{2\lambda(1 + \beta_{11})} + \frac{\beta_{11}}{1 + \beta_{11}}W, \quad (5)$$

which equals $W/2$ at $\beta_{11} = 1$.

I define two terms used throughout: *frontier effort* and *tied-state value*.

Definition 1 (Frontier effort). *The frontier effort is the symmetric equilibrium effort x_{11}^* in the development state.*

Both firms have reached the technological frontier, each one breakthrough from the prize, so x_{11}^* measures how hard the frontier itself is being pushed.

Definition 2 (Tied-state value). *The tied-state value is the equilibrium value V_{11} of competing in the development state.*

It is the value of arriving at the frontier tied rather than ahead or behind.

Comparative statics. Write (3) as $F(x; \theta) = 0$ for $\theta \in \{W, c, \lambda, \beta_{11}\}$. By the Implicit Function Theorem, $\partial x^*/\partial \theta = -F_\theta/F_x$, and at the subtracted root $F_x(x^*) = -\sqrt{\Delta} < 0$ exactly, so each sign is simply the sign of the corresponding partial F_θ (Appendix B).

Two of the four signs are clear. A larger prize raises effort: $F_W = \lambda(1 - \beta_{11})(2 - Mx) > 0$. A higher effort cost lowers it: $F_c = x(Mx - 3) < 0$.

The economic intuition behind the development rate λ is less obvious, because it scales both firms' success probabilities at once: a higher λ raises the return to own effort but also the threat of being pre-empted. The direct effect wins, and effort rises: $F_\lambda = x(3c - u)/\lambda > 0$, since under (A1)–(A2) we have $u < c/2 < 3c$. No auxiliary condition (such as $12c > W$) is required.

The spillover β_{11} lowers effort through free-riding: a fraction of the rival's effort now arrives in one's own success probability at no cost, so own effort and rival effort become partial substitutes and each firm supplies less of its own (Appendix B).

Proposition 1 (Frontier effort). *Under (A1)–(A2), the symmetric development effort x_{11}^* is strictly increasing in W and λ , and strictly decreasing in c and β_{11} .*

The comparative statics of the value function follow from the key inequality $\sqrt{\Delta} > 3c - u$, which holds because $\Delta - (3c - u)^2 = 4cu > 0$ (Appendix B):

Proposition 2 (Tied-state value). *Under (A1)–(A2), V_{11} is strictly increasing in β_{11} , W , and c , and strictly decreasing in λ .*

The value statics matter more than the effort statics, as the next two subsections show. V_{11} is the payoff the laggard chases in the asymmetric states, and the payoff to a simultaneous breakthrough in the initial state. Whatever moves V_{11} therefore spills backward into every earlier state of the race.

The result $\partial V_{11}/\partial \beta_{11} > 0$ is the *anticipatory catch-up channel*: a firm at the frontier shares in what its rival produces there, so a state with higher knowledge spillovers is more valuable. This is proven analytically (Appendix B). It is the value-side counterpart to the free-riding result in Proposition 1: higher β_{11} simultaneously depresses frontier effort and raises the value of reaching the frontier.

The sign $\partial V_{11}/\partial c > 0$ is less obvious: a higher cost parameter softens the rival's equilibrium effort by more than it raises own costs, so the tied state becomes more valuable.

Finally, V_{11} itself moves with c and λ . The shifts are small, about ± 2 –3% across the maintained region (Section 5), but they change how the comparative statics must be read: when c or λ moves, the asymmetric-state incentives change twice, once directly and once through V_{11} , so the next subsection must separate direct effects from total effects.

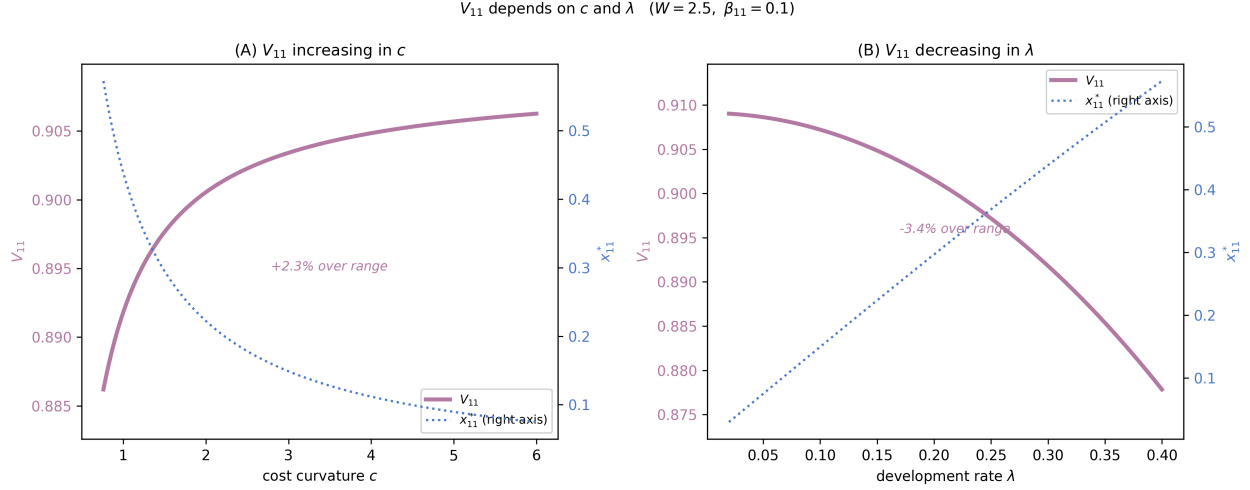


Figure 2: V_{11} is increasing in c and decreasing in λ (Proposition 2). The magnitudes are modest, but the dependence is what forces the total-effect decomposition in Section 4.2.

4.2 Asymmetric States $(1, 0)/(0, 1)$

Denote the leader's effort by z with breakthrough probability $P_L = \lambda z$, and the laggard's effort by x with breakthrough probability $P_F = \mu x$. There are no spillovers in this state, and the leader's development breakthrough takes precedence in the event of simultaneous success. Because $\beta = 0$ here, the laggard draws no contemporaneous benefit from the leader's effort; its only link to the frontier's spillovers is the value V_{11} of the tied state it is racing toward. Every effect of β_{11} on laggard behavior must therefore run through V_{11} . The Bellman equations are:

$$V_{10} = -\frac{c}{2}z^2 + P_L W + (1 - P_L)P_F V_{11} + (1 - P_L)(1 - P_F)V_{10}, \quad (6)$$

$$V_{01} = -\frac{c}{2}x^2 + (1 - P_L)P_F V_{11} + (1 - P_L)(1 - P_F)V_{01}. \quad (7)$$

The first-order conditions are:

$$\text{Leader:} \quad cz = \lambda(W - V_{10} + \mu x(V_{10} - V_{11})), \quad (8)$$

$$\text{Laggard:} \quad cx = \mu(1 - \lambda z)(V_{11} - V_{01}). \quad (9)$$

Substituting the value functions out of the FOCs yields a pair of best-response quadratics (Appendix C):

$$G_L = \frac{c\lambda}{2}(1 - \mu x) z^2 + c\mu x z - \lambda\mu x(W - V_{11}) = 0, \quad (10)$$

$$G_F = \frac{c\mu}{2}(1 - \lambda z) x^2 + c\lambda z x - \mu\lambda z(1 - \lambda z)V_{11} = 0. \quad (11)$$

In each, the constant and leading coefficients have opposite signs ($C/A < 0$), so the additive root is the unique positive root. The equilibrium (z^*, x^*) is the fixed point of these best responses. It has no convenient closed form, but it is transparent graphically: Figure 3 plots the two best-response curves, and the equilibrium is their crossing. The curves cross at most once (Appendix C.5), so an interior fixed point, where one exists, is unique. The crossing lies below the 45° line, meaning the leader out-invests the laggard, which is Proposition 3 below.

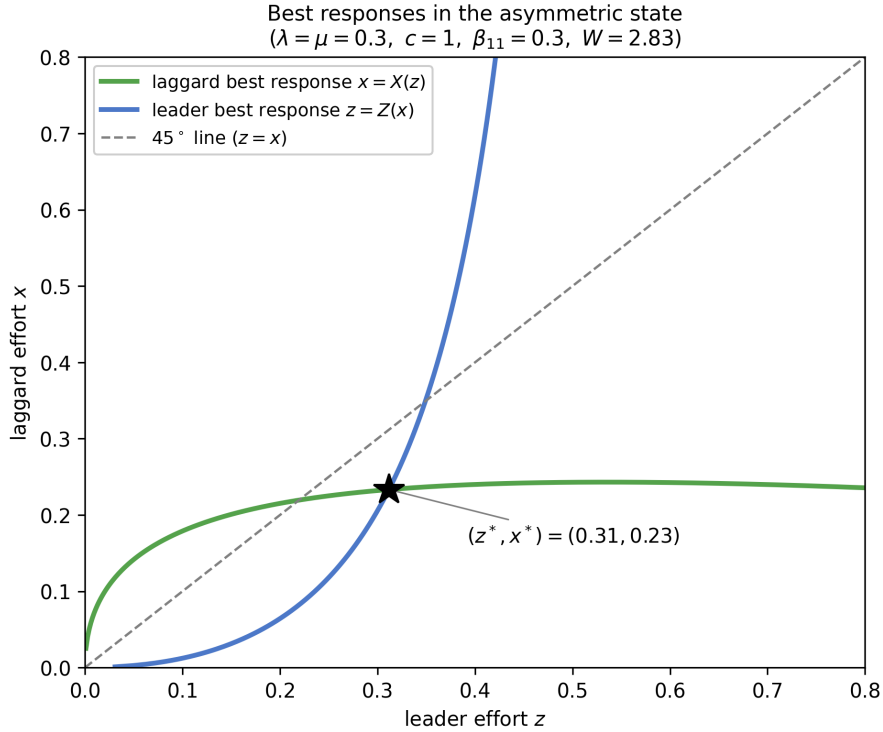


Figure 3: The asymmetric-state equilibrium as the crossing of the two best-response curves, at a representative parameter point with $\mu = \lambda$. The crossing below the 45° line is leader dominance (Proposition 3); the gentle hump in the laggard's curve is the non-monotonicity discussed below.

Lemma 3 (Value ordering). *Under (A1)–(A2), $0 \leq V_{01} \leq V_{11} < W/2 \leq W - V_{11}$, and $V_{11} \leq V_{10} \leq W$.*

The two key steps are exact: $V_{11} < W/2$ follows from evaluating the $(1, 1)$ quadratic at $x = \lambda W(1 - \beta_{11})/c$, which gives $F_{11} = -\lambda W(1 - \beta_{11}) < 0$; and $V_{10} \geq V_{11}$ follows from a mimicry argument, since $cx_{11}^* < \lambda(W - V_{11})$ (Appendix C). This lemma also provides the conditions needed for the root-selection result in state $(0, 0)$ below.

Proposition 3 (Leader dominance). *Assume the laggard’s research rate does not exceed the leader’s development rate ($\mu \leq \lambda$). Then in the asymmetric state, the leader strictly out-invests the laggard: $z^* > x^*$.*

The proof (Appendix C) works at the equilibrium rather than pointwise. Eliminating W and V_{11} from the two best-response quadratics yields an exact equilibrium identity relating the leader’s and laggard’s efforts; a purely algebraic inequality (independent of W and V_{11}) shows that this identity is incompatible with $x^* \geq z^*$ once $V_{11} < W/2$ (Lemma 3) and $\mu \leq \lambda$.

The hypothesis is one-sided: only $\mu > \lambda$ can overturn leader dominance. Numerically, leader dominance never fails for $\mu \leq \lambda$, and among $\mu \neq \lambda$ draws it fails in roughly 22%, with the failure share rising in μ/λ (from 0% at the boundary to 81% for $\mu/\lambda \geq 3$) (Section 5). The economic intuition: the leader’s effort advantage rests on the leader not being out-classed in raw research productivity. If the laggard’s catch-up technology is sufficiently more productive than the leader’s development technology, the ordering can reverse.

The laggard’s best response is non-monotone. Differentiating (11) along the best response gives an exact slope identity:

$$\text{sign}(X'(z)) = \text{sign}(\mu x(1 - \lambda z)^2 - 2\lambda^2 z^2). \quad (12)$$

The two arms have a clean economic reading, visible in the laggard’s first-order condition $cx = \mu(1 - \lambda z)(V_{11} - V_{01})$. A stronger leader does two things to the laggard. It erodes the

value of staying behind (V_{01} falls, so the gap $V_{11} - V_{01}$ widens and catching up becomes more urgent), and it raises the chance the race ends this period (the factor $(1 - \lambda z)$ shrinks, taxing the return to any effort at all). Below a threshold in z the first force dominates and the laggard fights harder (the *provocation* arm); above it the second takes over and the laggard retreats (the *discouragement* arm).

This distinction matters for how the discouragement effect should be read in this model. *Level* discouragement, the laggard being out-invested, is always present for $\mu \leq \lambda$ (Proposition 3). *Slope* discouragement, laggard effort falling in leader effort, is rare: numerically, the equilibrium sits on the provocation arm in 99.8–99.9% of the maintained region (Section 5). The model’s discouragement is a weakening of effort, not an exit from the race, and spillovers act on the level margin through V_{11} .

Transmission of β_{11} to the laggard. The result is a chain of three theorems, each holding for $\mu \leq \lambda$:

*a higher development-stage spillover raises the tied-state value (Proposition 2);
a higher tied-state value raises the laggard’s catch-up effort (Proposition 5); and
because the spillover reaches the asymmetric state only through that value, the
two steps combine — a higher development-stage spillover makes the laggard fight
harder (Corollary 1).*

The middle link is the one that binds: $dx^*/dV_{11} > 0$ holds whenever $x^* < \sqrt{2}z^*$, a condition strictly weaker than leader dominance and guaranteed for $\mu \leq \lambda$ by Proposition 3. Beyond the hypothesis, in the $\mu > \lambda$ region, the result is conditional on $x^* < \sqrt{2}z^*$, an observable equilibrium statistic; numerically, the condition holds at every scanned equilibrium in the maintained region, including every draw where leader dominance itself fails (Section 5.5). The leader’s response $dz^*/d\beta_{11}$ has an exact sign characterization (Lemma 8, Appendix C.5), but the sign itself is ambiguous: negative in roughly 95.5% of draws, positive in a structured high- λ /low- μ corner (Section 5.5).

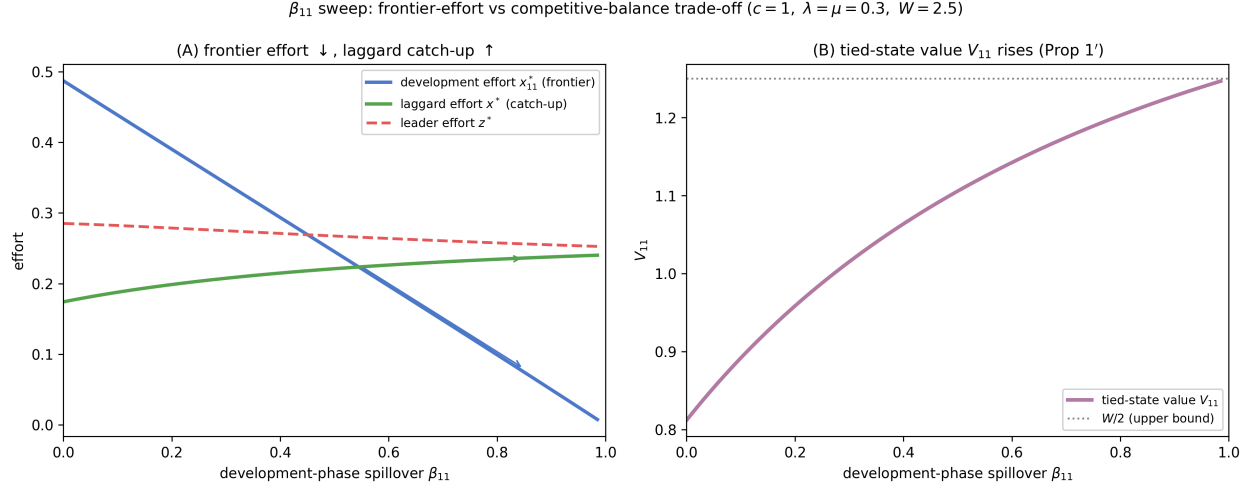


Figure 4: The central trade-off, swept in β_{11} at $\mu = \lambda$: frontier effort x_{11}^* falls (Proposition 1), the tied-state value V_{11} rises (Proposition 2), and the laggard's catch-up effort x^* rises (Corollary 1).

The leader's two channels. A richer tied state pulls the leader in two directions at once. The direct effect is negative: a higher V_{11} makes being caught less costly, so the leader's urgency to finish first falls. The strategic effect is positive: a higher V_{11} pushes the laggard to work harder, and the leader's best response is increasing in the laggard's effort. The exact boundary between the two regimes is the sign condition of Lemma 8, $\frac{1}{2}\lambda z^3(1 - \lambda z) \geq \mu(1 - \lambda z)x^3 + \lambda z x^2$: the strategic effect wins only when the leader's position is strong enough, which is the high- λ /low- μ corner where $dz^*/d\beta_{11} > 0$ in the scans (Section 5.5).

These positive-response cases are a small part of the region—about 4.5%—but they make economic sense. When development is fast and research slow (λ high, μ low), the leader almost always wins before the laggard can land the breakthrough it needs to catch up. A higher V_{11} helps the leader through just one event: being caught, which it makes less costly. When catch-up is unlikely, that direct channel barely operates. The strategic channel survives. A higher V_{11} pushes the laggard to work harder, by the same transmission result; and since the leader's best response rises with the laggard's effort ($Z' > 0$), a harder-working laggard raises the leader's own marginal return to effort. With the direct channel muted and the strategic one intact, the leader meets higher frontier spillovers with *more* effort. Lemma 8 draws the boundary exactly: a strong leader position—large λz , small μx —is what tips

it to the strategic term. This is the same corner where the indirect V_{11} channel reverses $dx^*/d\lambda$ (Section 5.5). Both come from one regime, where the leader's strategic response to the laggard, rather than direct payoff effects, drives the asymmetric-state comparative statics.

Total effects. Because V_{11} depends on c and λ in this discrete-time model, comparative statics with respect to these parameters combine a direct effect and an indirect effect through V_{11} :

$$\frac{d}{d\theta} = \frac{\partial}{\partial\theta} + \frac{\partial}{\partial V_{11}} \cdot \frac{\partial V_{11}}{\partial\theta}, \quad \theta \in \{c, \lambda\}. \quad (13)$$

Numerically, $dz^*/dc < 0$ and $dx^*/dc < 0$ hold in every draw, and $dz^*/d\lambda > 0$ in all but one of 8,000. The reversals concentrate in $dx^*/d\lambda$, which flips in roughly 2% of draws, all inside a high- λ /low- μ band where the indirect V_{11} channel overturns the direct sign (Section 5).

4.3 Initial Research State (0, 0)

Both firms are in the research phase, so spillovers are active: $P_i = \mu(x_i + \beta_{00}x_j)$. Four transitions are possible, including the direct jump to (1, 1) when both firms succeed simultaneously (probability P_iP_j). The Bellman equation for firm i is:

$$V_{00} = -\frac{c}{2}x_i^2 + P_i(1 - P_j)V_{10} + P_j(1 - P_i)V_{01} + P_iP_jV_{11} + (1 - P_i)(1 - P_j)V_{00}. \quad (14)$$

Let $m \equiv \mu(1 + \beta_{00})$, $D \equiv V_{10} - V_{01}$, and $G \equiv V_{10} + \beta_{00}V_{01} - (1 + \beta_{00})V_{11}$. Following the same steps as in state (1, 1), namely FOC, symmetry, and equating the FOC-implied and Bellman-implied values (Appendix D), the equilibrium reduces to the quadratic:

$$cmx^2 - [3c + 2\mu mG]x + 2\mu(1 - \beta_{00})D = 0, \quad (15)$$

solved by the subtracted root x_{00}^* .

Lemma 4 (Master quadratic). *Substituting $(V_{10}, V_{01}, V_{11}) \rightarrow (W, 0, W/2)$, $\mu \rightarrow \lambda$, and $\beta_{00} \rightarrow \beta_{11}$ into (15) recovers the (1, 1) quadratic (3) exactly.*

The same quadratic solves both symmetric states. The development state is the special case where the continuation values are the terminal payoffs. This is both a consistency check and a shortcut: the root-selection and comparative-statics methods carry over directly.

Lemma 5 (Root selection, $(0, 0)$). *Given the value ordering (Lemma 3) and $(A2')$, evaluating (15) at $x = 1/m$ gives $mF_{00}(1/m) = -2c - 2\mu m[\beta_{00}(V_{10} - V_{11}) - (V_{11} - V_{01})] < 0$, so the roots straddle $1/m$, both roots are positive, and the subtracted root is the unique effort consistent with valid probabilities.*

The inequality is not immediate from the display, since the bracketed term can take either sign; the proof bounds it using the value ordering of Lemma 3 and $(A2')$, via $mF_{00}(1/m) \leq -2c + 2\mu mW \leq -2c + 2mc < 0$ (Appendix D.3).

The discriminant Δ_0 of (15) is strictly positive analytically when either $m < \frac{1}{2}$ or $\beta_{00} \geq \frac{4}{5}$; elsewhere, positivity is confirmed numerically across the maintained region, with a minimum of approximately 0.02 and no violations in any scan (Section 5). Interiority $x_{00}^* \in (0, 1)$ likewise holds with no numerical violations.

In the rare-breakthrough limit, the equilibrium reduces to a familiar form:

$$x_{00}^* = \frac{2\mu(1 - \beta_{00})(V_{10} - V_{01})}{3c} + O(\mu^2), \quad (16)$$

so the simple proportional-to-the-value-gap formula from the mutually-exclusive-breakthrough model is recovered as the small- μ limit of the present model.

Proposition 4 (Early effort and continuation values). *Under $(A1)$ – $(A2')$, the equilibrium initial effort x_{00}^* is strictly increasing in V_{10} and V_{11} , and strictly decreasing in V_{01} .*

By the same IFT argument as before, each sign follows from the corresponding partial of (15) (Appendix D). Two are immediate: $F_{00, V_{01}} < 0$, and $F_{00, V_{10}} > 0$ holds unconditionally

under (A2') via the exact evaluation $F_{00}((1 - \beta_{00})/m) < 0$. The third is new relative to the mutually-exclusive-breakthrough model: $F_{00,V_{11}} = 2\mu m(1 + \beta_{00})x > 0$ operates entirely through the *direct-jump channel*, the $P_i P_j$ transition straight from $(0, 0)$ to $(1, 1)$. Without simultaneous breakthroughs the tied state is unreachable from the starting line, and V_{11} never enters the initial-state problem.

Because V_{11} rises with β_{11} (Proposition 2), frontier spillovers are felt before either firm reaches the frontier: two firms still in research push harder when the state they might jump to together is worth more. A higher β_{11} also raises V_{10} and V_{01} , so being ahead and being behind gain value at the same time, and the gap D that drives early effort can widen or narrow. Proposition 4 signs each partial separately; it does not sign the total effect of β_{11} on early effort.

5 Numerical Analysis

The theorems characterize behavior inside the maintained region; the scans show where its edges actually lie and how far the results reach beyond them. The section does two things. First, it runs one consolidated consistency check: every closed form and every proven result in Section 4 is verified against an independent numerical solution of the model. Because those results are theorems, the numerics add no evidence (they only guard against algebra mistakes), so the check is kept to a single passage. Second, and more importantly, it examines what remains numerical: where the maintained assumptions bind, where the hypothesis of Proposition 3 matters, how far the transmission theorem of Appendix C.5 reaches beyond the case in which it is proven, and which regularity properties hold only in the scans. Readers can skim the consistency check of Section 5.2 and turn to Sections 5.4 and 5.5, which hold the main findings.

5.1 Purpose and Method

The independent solver never touches the closed forms. Instead, it solves the Bellman equations and first-order conditions directly, by bisection in the symmetric states and best-response iteration in the asymmetric states, to an iteration tolerance of 10^{-13} . If the closed forms and the independent solver agree, the algebra in Section 4 and the Appendix is correct.

The checks come in three layers. Every algebraic identity in the appendices — the master-quadratic substitution, the derivative identity behind Proposition 2, the best-response factorizations, the cross-partials, the Jacobian determinant, and the Cramer numerators — is verified in SymPy and required to simplify to exact zero, so those claims rest on no tolerance. Comparative-statics signs are checked by central finite differences at solved equilibria. A finite difference near zero cannot resolve the sign, so I re-solve every flagged case in mpmath at 50 digits; every apparent sign violation in the scans came from float resolution, and all of them cleared (Section 5.2).

All scans draw random parameters inside the maintained region (A1)–(A2′). I use fixed random seeds and report sample sizes with every result so that all checks can be replicated. The core scans use seeds 11, 21, and 31 with 60,000, 40,000, and 30,000 draws. The leader-dominance frontier of Section 5.4 uses seed 101 with 200,000 draws. The transmission scans of Section 5.5 use seed 41 with 60,000 draws, and seed 43 with 60,000 for the margin probe of the $\sqrt{2}$ condition. The Jacobian and Cramer identities are checked at 20,000 solved equilibria under seed 8, and the confirmation re-runs of Section 5.2 use seeds 202 and 707 with 60,000 draws each. Every reported solution satisfies a residual tolerance of 10^{-9} . The full scripts and result logs are saved alongside the thesis for replication.⁵

⁵Every scan reported in this section, with its seed, sample size, and raw output log, is in the replication repository: https://github.com/iellahi/Honors_Thesis

5.2 Consistency and Robustness Checks

Every check passes. The closed-form effort x_{11}^* from Equation (4) matches the independent solver to within 3.5×10^{-14} across 60,000 draws, with no violations of interiority or of valid probabilities. Best-response iteration in the asymmetric state converges in every draw, and the resulting (z^*, x^*) satisfies the quadratics (10) and (11) with residuals no larger than 9.5×10^{-15} . The value ordering of Lemma 3 holds with zero violations, and Proposition 3 holds in every draw satisfying its hypothesis $\mu \leq \lambda$ (including all $\mu = \lambda$ draws), exactly as the extended proof requires. The master-quadratic identity of Lemma 4 simplifies to zero symbolically. The Jacobian and transmission formulas of Appendix C.5 were likewise re-derived symbolically and then cross-checked at 20,000 solved equilibria, where they agree with finite-difference derivatives to a relative error below 10^{-5} . Finally, the proven comparative-statics signs (Propositions 1, 2, and 4) hold in every draw once near-zero derivatives are re-evaluated at 50-digit precision; every flagged case was a finite-difference artifact, with zero real violations. The scan also confirms the fact from Section 4.1 that V_{11} moves with c and λ , by about ± 2 –3% across the maintained region (Figure 2), which is what forces the total-effect decomposition in Section 4.2. As a final pass, I re-ran the entire battery on two fresh seeds (202 and 707, 60,000 draws each). Both reproduce every result. They flagged 239 and 236 cases, all falling in two slots: $\partial V_{11}/\partial c$ (Proposition 2), where V_{11} sits near its large- c saturation and the true derivative falls below float finite-difference resolution, and $\partial x_{00}^*/\partial V_{11}$ (Proposition 4) at very small μ , where $F_{00,V_{11}} = 2\mu m(1 + \beta_{00})x$ is of order μ^2 . Re-evaluating each flagged case at 50-digit precision recovers the proven sign in every one. No numerical claim produced a real violation.

5.3 Where the Assumptions Bind

The corner regime and (A2). Assumption (A2) is sufficient but not tight. At the boundary $\lambda W = c$, the interior root is only about 0.63–0.66, well inside $(0, 1)$. Pushing beyond the assumption, the true switch from interior to corner equilibria occurs around

$\lambda W/c \approx 1.53\text{--}1.66$ (Figure 5). So interior equilibria survive in a band beyond (A2), and the assumption guarantees the analytical results rather than marking the true edge of the interior regime. Finally, in a band with (A2) relaxed, 19.8% of draws land in the corner.

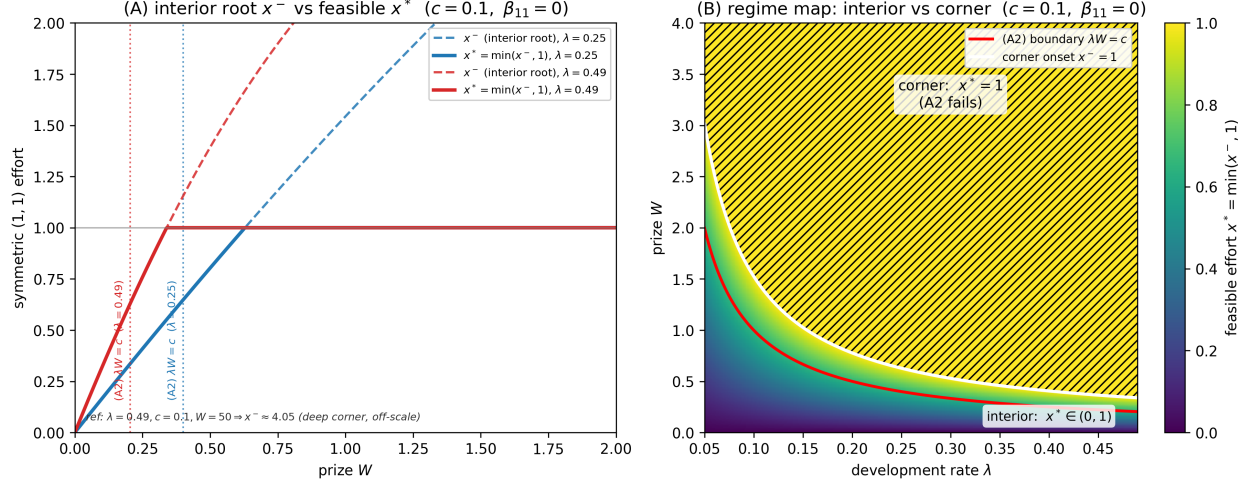


Figure 5: The interior/corner frontier. (A2) requires $\lambda W \leq c$, but interior equilibria persist until $\lambda W/c \approx 1.5\text{--}1.7$.

The initial state and (A2'). The discriminant Δ_0 of the initial-state quadratic (15) is proven positive when either $m < \frac{1}{2}$ or $\beta_{00} \geq \frac{4}{5}$; elsewhere the claim is numerical, and this is exactly where the check matters. Δ_0 is positive in every draw across all scans, with a minimum of approximately 0.02 landing in the region the proof does not cover. Interiority holds throughout: $x_{00}^* \in (0, 1)$ and $m x_{00}^* < 1$ with zero violations, and the sharper bound $m x_{00}^* < 1 - \beta_{00}$, the numerical counterpart of the exact evaluation behind the unconditional sign $\partial x_{00}^* / \partial V_{10} > 0$ in Proposition 4, holds in all 30,000 draws tested (Figure 7). Relaxing (A2') itself produces no interiority violations at all: the assumption buys the analytical proofs, not the good behavior of the equilibrium. Figure 6 sweeps the initial-state objects in β_{00} . Two features stand out. The breakthrough probability is nearly flat at low β_{00} (diffusion substitutes for own effort roughly one-for-one before free-riding takes over), and $x_{00}^* = 0$ exactly at $\beta_{00} = 1$, the master-quadratic analogue of the (1,1) result. Yet V_{00} rises in β_{00} : ex ante, both firms prefer a starting line with higher spillovers. That observation is numerical rather than proven.

β_{00} sweep ($c = 1$, $\lambda = \mu = 0.3$, $W = 2.5$, $\beta_{11} = 0.3$)

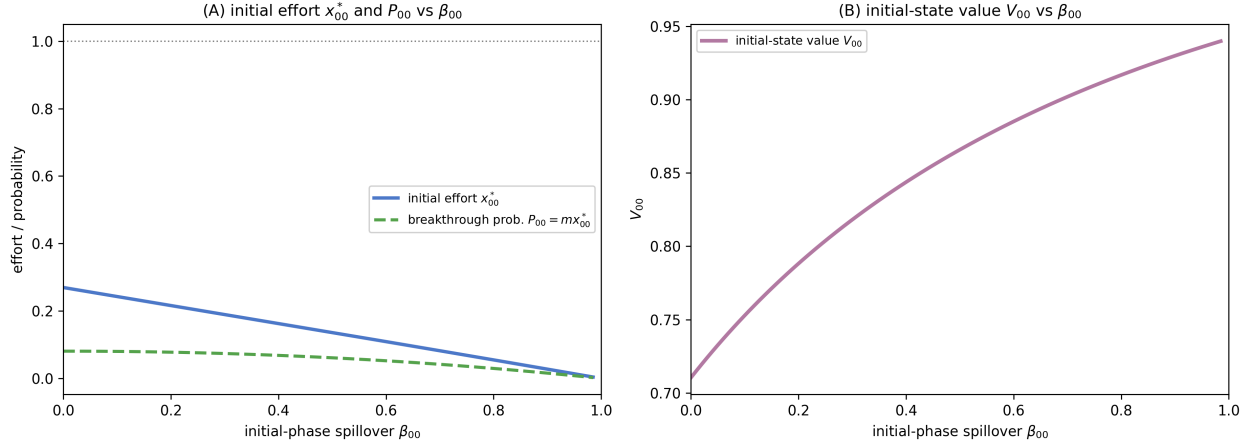


Figure 6: Baseline sweep of the initial-state equilibrium objects as β_{00} varies: effort and the breakthrough probability fall (the latter only once free-riding outweighs diffusion), while the state value V_{00} rises.

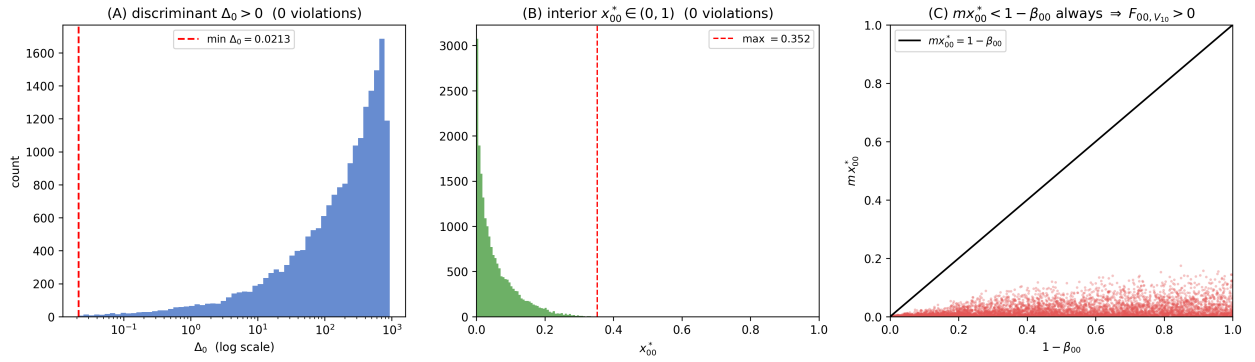


Figure 7: Initial-state regularity across the scan. (A) Δ_0 stays positive, with its minimum of 0.021 falling in the region the analytical proof does not cover. (B) x_{00}^* is interior in every draw. (C) $mx_{00}^* < 1 - \beta_{00}$ always, which is what makes the sign $\partial x_{00}^* / \partial V_{10} > 0$ in Proposition 4 unconditional.

5.4 The Limits of Leader Dominance

Proposition 3 states that the leader out-invests the laggard when $\mu \leq \lambda$. Throughout that region the result holds in every draw. Beyond the hypothesis, for $\mu > \lambda$, the picture changes sharply: across $\mu \neq \lambda$ draws leader dominance fails in 22.0% overall (seed 101, 200,000 draws), with every failure occurring at $\mu > \lambda$, and the failure share rises steadily in the ratio μ/λ , from 0% whenever $\mu \leq \lambda$ to 81% once $\mu/\lambda \geq 3$ (Figure 8). The economic reading is the one given in Section 4.2: the leader's effort advantage rests on the leader not being out-classed in raw research productivity. The numerics map exactly where the proposition holds

and where it does not. What they do *not* show is the transmission mechanism failing there: as the next subsection shows, the transmission result survives even where leader dominance does not.

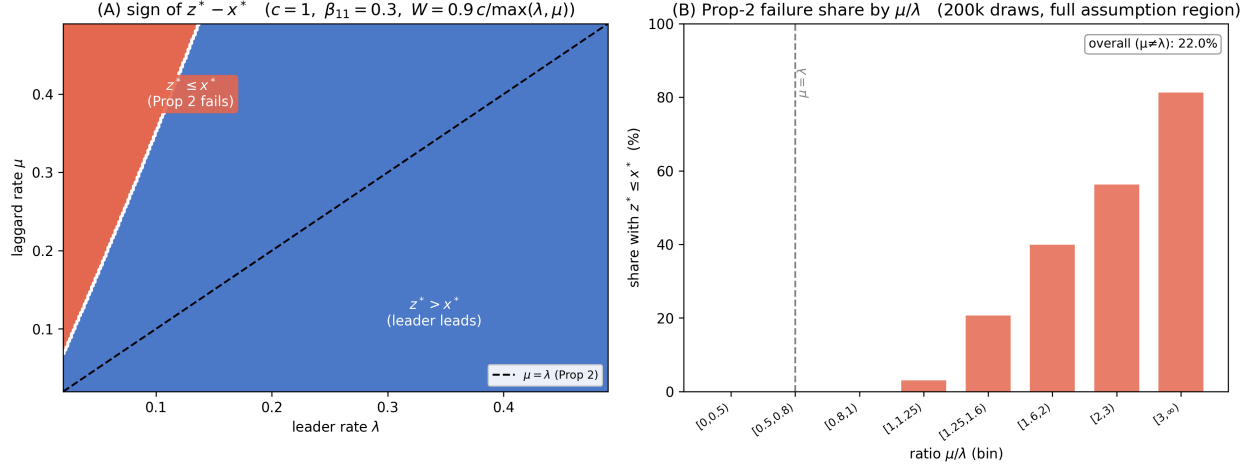


Figure 8: The leader-dominance frontier (Proposition 3). Leader dominance ($z^* > x^*$) never fails for $\mu \leq \lambda$ and fails increasingly often as μ/λ grows.

5.5 Transmission: The Theorem's Reach Beyond $\mu \leq \lambda$

For $\mu \leq \lambda$, the transmission of development-stage spillovers to the laggard, $dx^*/d\beta_{11} > 0$, is a theorem (Corollary 1, Appendix C.5). The numerics therefore map how far the theorem reaches beyond its hypothesis, into the $\mu > \lambda$ region.

The reach of the $\sqrt{2}$ condition. The proven sufficient condition is $x^* < \sqrt{2} z^*$ (Proposition 5), which is strictly weaker than leader dominance. In a separate 60,000-draw scan (seed 41) across the $\mu \neq \lambda$ region, the condition held in every one, including all 13,176 draws in which leader dominance itself fails. Transmission is therefore proven at every equilibrium the scans can reach, even where the leader is out-invested. The margin, however, is not uniform. Figure 9 maps the ratio $x^*/(\sqrt{2} z^*)$: it stays below one everywhere but approaches one in the corner where $\lambda \rightarrow 0$, $\beta_{11} \rightarrow 1$, and $\mu \rightarrow \frac{1}{2}$, without attaining it. $\sqrt{2}$ is where the third bracketed term of the transmission numerator changes sign, so it is the best constant this argument gives; the other two terms stay positive, so it is possible that transmission

survives past it. It is therefore a conditional theorem with a region-wide numerical check of the condition, not an unconditional claim.

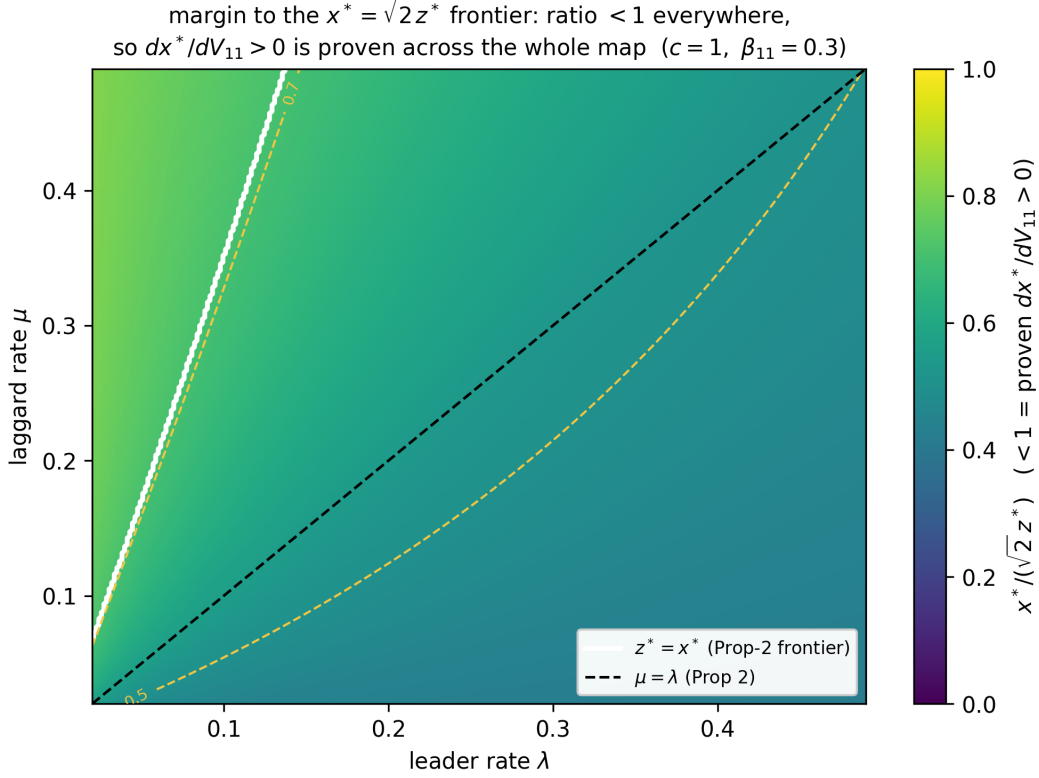


Figure 9: The margin to the transmission frontier on the grid of Section 5.4. The ratio $x^*/(\sqrt{2}z^*)$ peaks at 0.81 here, including where Proposition 3 fails (above the white curve), so $dx^*/dV_{11} > 0$ is proven at every equilibrium shown. A separate corner sweep pushes the ratio to 0.99 without reaching one, so the hypothesis is nearly binding there; whether transmission survives past $\sqrt{2}$ is untested.

The leader’s response. The sign of $dz^*/d\beta_{11}$ is ambiguous, but its boundary is now exact: the sign equals the sign of $\frac{1}{2}\lambda z^3(1 - \lambda z) - \mu(1 - \lambda z)x^3 - \lambda z x^2$ (Lemma 8), and this analytical boundary tracks the numerical sign flip precisely, with agreement at all 40,000 scanned equilibria (Figure 10). The leader’s effort falls in β_{11} in roughly 95.5% of draws and rises only in a high- λ /low- μ corner; the economic reading of that corner—why a nearly-safe leader’s strategic response to a provoked laggard outweighs its direct complacency motive—is developed in Section 4.2. I make no general claim about this sign.

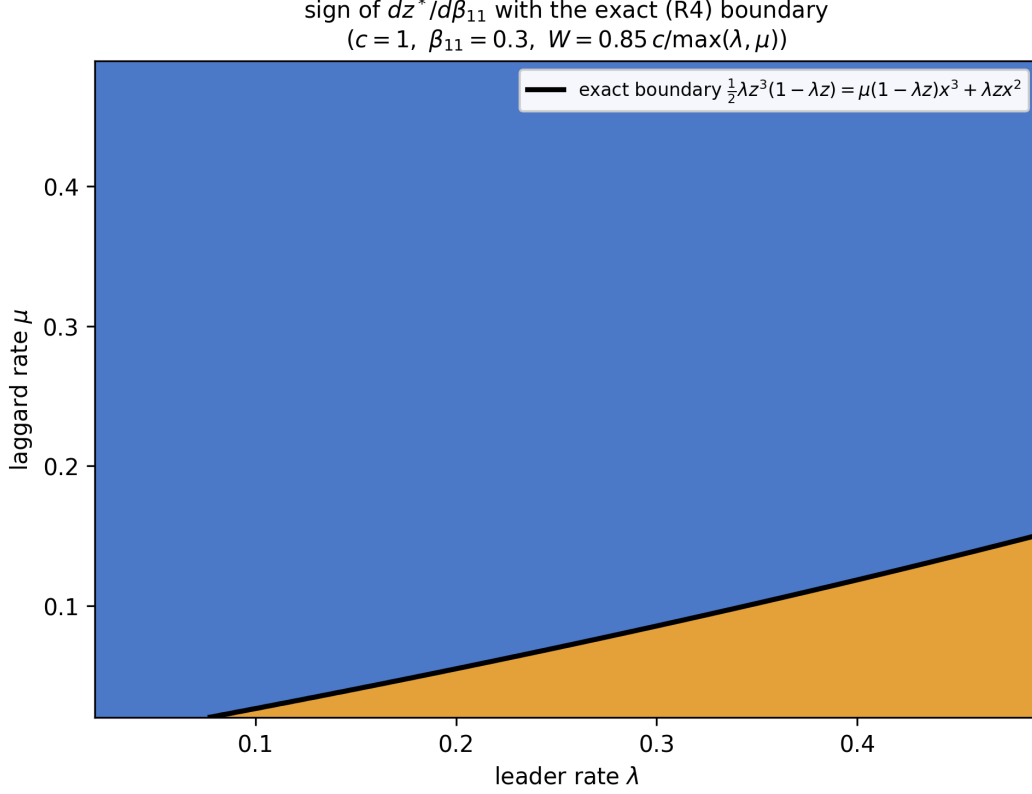


Figure 10: Sign map of $dz^*/d\beta_{11}$: negative (blue) in most of the region, positive (orange) in the high- λ /low- μ corner. The black curve is the exact boundary $\frac{1}{2}\lambda z^3(1 - \lambda z) = \mu(1 - \lambda z)x^3 + \lambda z x^2$ from Lemma 8, not a fitted line. The panel is one (c, β_{11}, W) slice; across the full region the positive share is 4.5%.

The total effect $dx^*/d\lambda$. The direct and indirect channels usually push the same way, but reversals occur in roughly 2% of draws (1.8–2.0% across seeds), concentrated in the same high- λ /low- μ corner, where the indirect channel through V_{11} flips the direct sign (Figure 11). The other total effects are close to uniform: $dz^*/dc < 0$ and $dx^*/dc < 0$ hold in every draw, and $dz^*/d\lambda > 0$ in 7,999 of 8,000.

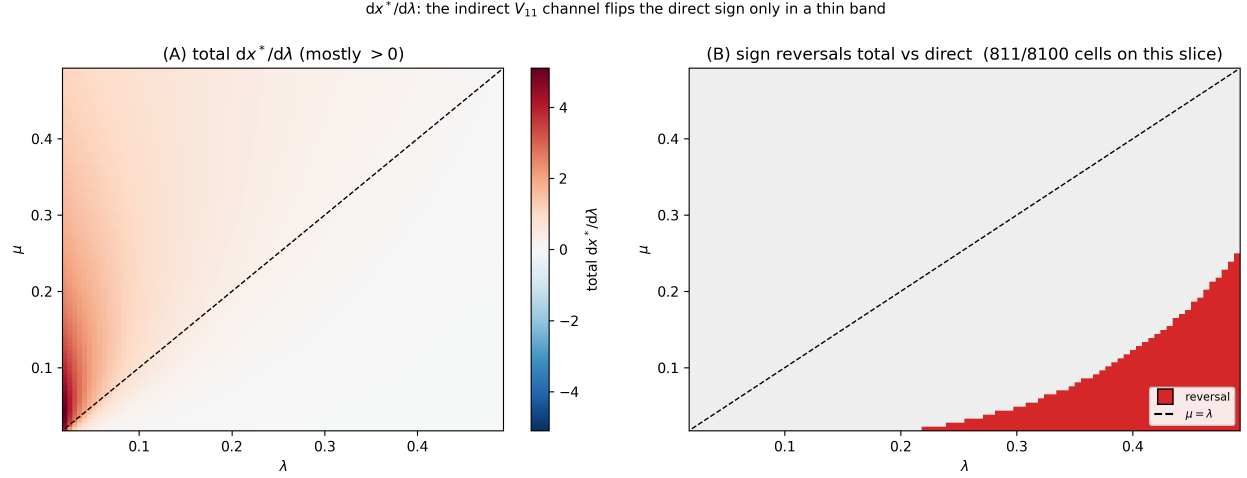


Figure 11: Reversals of $dx^*/d\lambda$ form a structured corner wedge rather than scattered noise. The map shows a fixed- (c, β_{11}, W) slice on which the wedge covers about 10% of the panel; the share across the full maintained region is 1.8–2.0%.

The two exceptions above are the same place. In the high- λ /low- μ regime (fast development, slow research), the leader is nearly certain to finish before the laggard’s next breakthrough arrives, and the equilibrium is dominated by the leader’s *strategic* response to the laggard rather than by direct payoff effects. That is where the leader’s spillover response flips sign, and where the indirect V_{11} channel is strong enough to reverse $dx^*/d\lambda$. Outside this one corner, every comparative static in the asymmetric state behaves as the direct effects predict.

5.6 Regularity and Open Questions

The equilibrium sits on the provocation arm of the laggard’s best response, where the laggard responds to a stronger leader by fighting harder, in 99.8–99.9% of the region (Figure 12). Slope discouragement is therefore rare in equilibrium, which is why Section 4.2 frames the model’s discouragement effect as a level effect. With $\det J > 0$ and uniqueness now proven (Appendix C.5), two claims remain numerical: the *existence* of an interior fixed point, for the reason given there, and the convergence of best-response iteration to it. Both held in every draw, which supports but does not prove them.

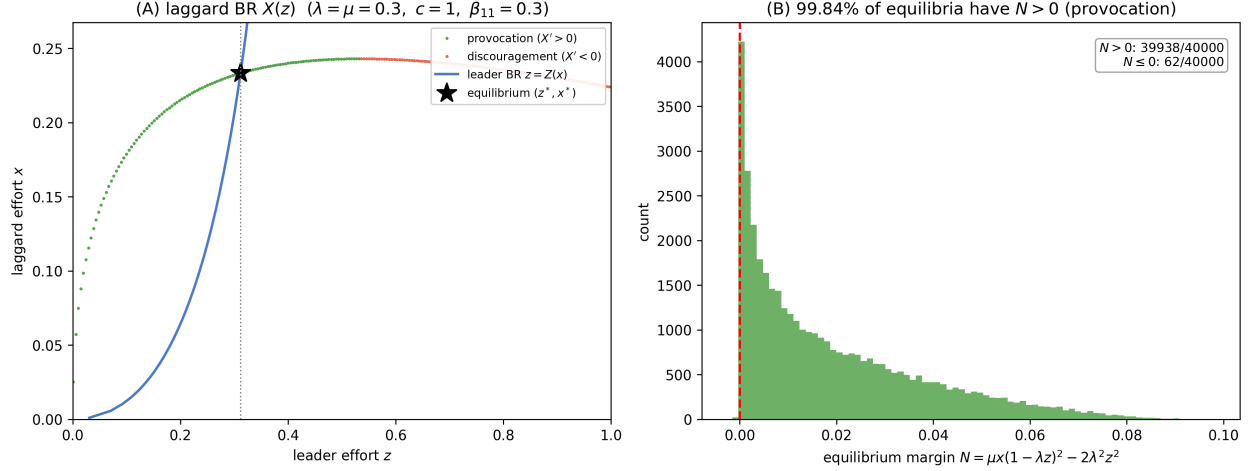


Figure 12: The laggard's non-monotone best response: a rising provocation arm followed by a falling discouragement arm. Equilibria almost always sit on the provocation arm.

Within the maintained assumptions, then, no numerical claim breaks and every theorem passes its consistency check, while each maintained assumption marks a real boundary: the corner regime beyond (A2), and the failure of leader dominance once $\mu > \lambda$ (Sections 5.3 and 5.4).

6 Conclusion

I asked what inevitable knowledge spillovers do to a winner-take-all innovation race with two firms, two stages, a prize for finishing first, and a spillover that operates only between firms at the same technological stage. Spillovers in the symmetric state depress frontier effort, because each firm free-rides on what spills over from its rival. The same spillovers raise the value of being tied at the frontier, and that value is what incentivizes the laggard to fight harder. Early-stage effort rises for the same reason.

The mechanism behind this is indirect. The laggard receives no contemporaneous benefit from the leader's effort; spillovers are inactive when the race is asymmetric. The laggard responds to the anticipated value of the state it is racing toward. The tied-state value rises with the spillover (Proposition 2), the laggard's effort rises with the tied-state value

(Proposition 5), and the chain closes whenever the laggard's research rate does not exceed the leader's development rate (Corollary 1). That last step needs only $x^* < \sqrt{2} z^*$, which holds at every equilibrium in the scans, so the result is a theorem well beyond the region where I can prove its hypothesis.

The model has many limitations. There are only two firms, and the spillover is exogenous; firms neither invest in secrecy nor in absorptive capacity. Furthermore, within-stage absorption is a linear approximation, not a claim about how absorption scales. The contest success function is linear, the prize is winner-take-all, and there is no discounting, which shuts off some sources of discouragement.

Each limitation points at the next step. The most natural is to make spillover a choice: the internal-tournament reading of the model puts β in the hands of a designer, and the trade-off documented here, frontier effort against competitive balance, is exactly the margin such a designer would price. Beyond that, an unconditional treatment of the transmission condition at unequal rates, endogenous secrecy or absorption investments, more than two firms, and prize structures softer than winner-take-all are all well-defined extensions of the same framework. The core mechanism, that knowledge spillovers affect the race through the value of the states they exist in, is likely to work in all of them.

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Appendix

A Symmetric Development State: Full Derivation

A.1 From the Bellman equation to the equilibrium quadratic

Bellman equation in state $(1, 1)$ with $P_i = \lambda(x_i + \beta_{11}x_j)$, $P_j = \lambda(x_j + \beta_{11}x_i)$:

$$V_{11} = -\frac{c}{2}x_i^2 + P_i(1 - P_j)W + P_iP_j\frac{W}{2} + (1 - P_i)(1 - P_j)V_{11}.$$

Expanding $(1 - P_i)(1 - P_j) = 1 - P_i - P_j + P_iP_j$, subtracting V_{11} , and collecting:

$$(P_i + P_j - P_iP_j)V_{11} = P_iW - P_iP_j\frac{W}{2} - \frac{c}{2}x_i^2 \implies V_{11} = \frac{W(2P_i - P_iP_j) - cx_i^2}{2(P_i + P_j - P_iP_j)}.$$

First-order condition. Differentiating the Bellman right-hand side in x_i , holding the continuation value V_{11} fixed, with $\partial P_i / \partial x_i = \lambda$, $\partial P_j / \partial x_i = \beta_{11}\lambda$, $\partial(P_iP_j) / \partial x_i = \lambda P_j + \beta_{11}\lambda P_i$:

$$cx_i = \lambda \left[W - (P_j + \beta_{11}P_i)\frac{W}{2} - (1 + \beta_{11})V_{11} + (P_j + \beta_{11}P_i)V_{11} \right].$$

Impose symmetry $x_i = x_j = x$, so $P_i = P_j = P = Mx$ with $M \equiv \lambda(1 + \beta_{11})$:

$$cx = \lambda W - MV_{11} - M^2x\left(\frac{W}{2} - V_{11}\right) \implies V_{11} = \frac{cx - \lambda W + M^2x\frac{W}{2}}{M(Mx - 1)}. \quad (\text{FOC-form})$$

The symmetric Bellman-form value is

$$V_{11} = \frac{W(2M - M^2x) - cx}{2M(2 - Mx)}. \quad (\text{Bellman-form})$$

Equating FOC-form and Bellman-form and clearing denominators gives, after simplification,

$$(Mc)x^2 - (3c + 2\lambda MW - M^2W)x + 2W(2\lambda - M) = 0,$$

and substituting $M = \lambda(1 + \beta_{11})$:

$$c\lambda(1 + \beta_{11})x^2 - (3c + \lambda^2W(1 - \beta_{11}^2))x + 2\lambda W(1 - \beta_{11}) = 0.$$

A.2 Existence of real roots

Lemma (1, Discriminant, restated). *Let $u \equiv \lambda^2W(1 - \beta_{11}^2) \geq 0$. The discriminant of the equilibrium quadratic is*

$$\Delta = 9c^2 - 2cu + u^2 = (u - c)^2 + 8c^2 > 0.$$

Hence real roots exist for all admissible parameters.

A.3 Root selection and interiority

Lemma (2, Root selection and interiority, (1, 1), restated). *Write the quadratic as $F(x) = Ax^2 + Bx + C$ with $A = c\lambda(1 + \beta_{11}) > 0$, $B = -(3c + \lambda^2W(1 - \beta_{11}^2))$, $C = 2\lambda W(1 - \beta_{11}) \geq 0$. Under (A1)–(A2) and $\beta_{11} < 1$:*

1. *Both roots are strictly positive: product $C/A > 0$ and sum $-B/A > 0$.*
2. *Probability validity. $M F(1/M) = \lambda^2W(1 - \beta_{11}^2) - 2c$. The chain*

$$\lambda^2W(1 - \beta_{11}^2) = \lambda(\lambda W)(1 - \beta_{11}^2) \leq \lambda c(1 - \beta_{11}^2) < \frac{c}{2} < 2c$$

(first inequality by (A2), second by $\lambda < \frac{1}{2}$ and $1 - \beta_{11}^2 \leq 1$) gives $F(1/M) < 0$. Since $A > 0$, the roots straddle $1/M$: $x^- < 1/M < x^+$. Thus the additive root yields $P = Mx^+ > 1$ (infeasible) and the subtracted root yields $P = Mx^- < 1$ (valid).

3. *Interiority. $F(1) = c(M - 3) + \lambda W(1 - \beta_{11})(2 - M)$. By (A2),*

$$\lambda W(1 - \beta_{11})(2 - M) \leq c(1 - \beta_{11})(2 - M),$$

and $c(3 - M) - c(1 - \beta_{11})(2 - M) = c(1 + \beta_{11}(2 - M)) > 0$, so $F(1) < 0$ and hence $x^- < 1 < x^+$. The interior bound $x_{11}^* = x^- \in (0, 1)$ holds.

Remark. Absent (A2) the interior solution can fail: e.g., $\lambda = 0.49$, $c = 0.1$, $W = 50$ gives $x^- \approx 4.05 > 1$, forcing a corner at $x^* = 1$.

A.4 Equilibrium effort and value

The valid (subtracted) root of the quadratic is

$$x_{11}^* = \frac{3c + \lambda^2 W(1 - \beta_{11}^2) - \sqrt{\Delta}}{2c\lambda(1 + \beta_{11})}, \quad \Delta = 9c^2 - 2c\lambda^2 W(1 - \beta_{11}^2) + \lambda^4 W^2(1 - \beta_{11}^2)^2.$$

Substituting x_{11}^* into the FOC-form value (equivalently the Bellman-form; the two agree, verified symbolically) yields

$$V_{11} = \frac{c x_{11}^*}{2\lambda(1 + \beta_{11})} + \frac{\beta_{11}}{1 + \beta_{11}} W.$$

At $\beta_{11} = 1$: $C = 0$, so $x_{11}^* = 0$ (full free-riding) and $V_{11} = W/2$.

B Comparative Statics in the Development State

Throughout, $u \equiv \lambda^2 W(1 - \beta_{11}^2)$, $M \equiv \lambda(1 + \beta_{11})$, and $F(x) = Ax^2 + Bx + C$ is the equilibrium quadratic with $A = cM$, $B = -(3c + u)$, $C = 2\lambda W(1 - \beta_{11})$, discriminant $\Delta = 9c^2 - 2cu + u^2$. Under (A1)–(A2), $u \leq \lambda c(1 - \beta_{11}^2) < c/2$.

B.1 The IFT denominator

At the subtracted root $x^* = (-B - \sqrt{\Delta})/(2A)$: $F_x(x^*) = 2Ax^* + B = -\sqrt{\Delta} < 0$. Hence by the Implicit Function Theorem, for any parameter θ ,

$$\frac{dx^*}{d\theta} = -\frac{F_\theta(x^*)}{F_x(x^*)} = \frac{F_\theta(x^*)}{\sqrt{\Delta}}, \quad \text{sign}\left(\frac{dx^*}{d\theta}\right) = \text{sign}(F_\theta(x^*)).$$

B.2 Effort comparative statics

W : $F_W = \lambda(1 - \beta_{11})(2 - Mx)$. Since $P = Mx^* < 1 < 2$: $F_W(x^*) > 0 \Rightarrow \partial x^*/\partial W > 0$.

c : $F_c = x(Mx - 3)$. Since $Mx^* < 1 < 3$: $F_c(x^*) < 0 \Rightarrow \partial x^*/\partial c < 0$.

λ : Polynomial division gives $F_\lambda = \frac{1}{\lambda}F + \frac{x(3c-u)}{\lambda}$, so at the root $F_\lambda(x^*) = x^*(3c - u)/\lambda$.

Under (A2), $u < c/2 < 3c$, hence $\partial x^*/\partial \lambda > 0$. (This holds without any auxiliary condition such as $12c > W$.)

β_{11} : Polynomial division gives $F_\beta(x^*) = [x^*(3c + M^2W) - 4\lambda W]/(1 + \beta_{11})$. Let $\bar{x} = 4\lambda W/(3c + M^2W)$; then $F_\beta(x^*) < 0 \iff x^* < \bar{x}$, which holds iff $F(\bar{x}) < 0$ (roots straddle $\bar{x}$). Direct computation:

$$F(\bar{x}) \frac{(3c + M^2W)^2}{\lambda W} = -18c^2(1 + \beta_{11}) + 4cM\lambda W(1 - 3\beta_{11}^2) - 2M^3\lambda W^2(1 - \beta_{11})^2.$$

Under (A1)–(A2): $4cM\lambda W(1 - 3\beta_{11}^2) \leq 4c^2M < 2c^2(1 + \beta_{11})$, so the right side is below $-16c^2(1 + \beta_{11}) < 0$. Hence $\partial x^*/\partial \beta_{11} < 0$. $\square$

B.3 Key inequality

Lemma 6 (Key inequality). $\sqrt{\Delta} > 3c - u$, *strictly, whenever $u > 0$; indeed $\Delta - (3c - u)^2 = 4cu$.*

B.4 Value comparative statics

Recall $V_{11} = \frac{cx^*}{2\lambda(1 + \beta_{11})} + \frac{\beta_{11}}{1 + \beta_{11}}W$ and $cMx^* = \frac{1}{2}(3c + u - \sqrt{\Delta})$.

$$c: \frac{\partial V_{11}}{\partial c} = \frac{x^*(\sqrt{\Delta} + u - 3c)}{4\lambda(1 + \beta_{11})\sqrt{\Delta}} > 0 \text{ by B.3 (and } x^* > 0 \text{ for } \beta_{11} < 1).$$

$$\lambda: \frac{\partial V_{11}}{\partial \lambda} = \frac{cx^*[(3c - u) - \sqrt{\Delta}]}{2\lambda^2(1 + \beta_{11})\sqrt{\Delta}} < 0 \text{ by B.3.}$$

$$W: \frac{\partial V_{11}}{\partial W} = \frac{c}{2\lambda(1 + \beta_{11})} \frac{\partial x^*}{\partial W} + \frac{\beta_{11}}{1 + \beta_{11}} > 0 \text{ (both terms non-negative, at least one strictly positive).}$$

β_{11} :

Proposition (2, anticipatory catch-up channel: the β_{11} sign of the tied-state value).

Under (A1)–(A2), $\partial V_{11}/\partial \beta_{11} > 0$.

Proof. Let $q \equiv \lambda^2 W/c$; (A1)–(A2) give $q = \lambda \cdot (\lambda W/c) < \frac{1}{2}$. Define

$$T_1 = 18 + 3q(1 + \beta_{11})^2 + q(1 - \beta_{11}^2) + q^2(1 - \beta_{11}^2)(1 + \beta_{11})^2 + q^2(1 - \beta_{11}^2)^2 - 8q(1 + \beta_{11}),$$

$$T_2 = 6 - 2q(1 + \beta_{11}), \quad \tilde{\Delta} = 9 - 2q(1 - \beta_{11}^2) + q^2(1 - \beta_{11}^2)^2.$$

Exact identity (verified symbolically):

$$\frac{\partial V_{11}}{\partial \beta_{11}} = \frac{c(T_1 - \sqrt{\tilde{\Delta}}T_2)}{4\lambda^2(1 + \beta_{11})^3\sqrt{\tilde{\Delta}}}.$$

For $q < \frac{1}{2}$, $\beta_{11} < 1$: (i) $T_2 > 6 - 2 \cdot \frac{1}{2} \cdot 2 = 4 > 0$; (ii) $T_1 > 18 - 8q(1 + \beta_{11}) > 10 > 0$;

(iii) the factorization

$$T_1^2 - \tilde{\Delta}T_2^2 = 16q(1 + \beta_{11})\left[q^2(1 - \beta_{11}^2)^2 - 2q(1 + \beta_{11}^3) + 9\right],$$

whose bracket exceeds $9 - 2 \cdot \frac{1}{2} \cdot 2 = 7 > 0$. Hence $T_1 > \sqrt{\tilde{\Delta}}T_2$ and $\partial V_{11}/\partial\beta_{11} > 0$. ■

C Asymmetric States: Derivations and Proofs

Throughout: $P_L = \lambda z$, $P_F = \mu x$, $S = P_L + P_F - P_L P_F$; V_{11} is the constant from Appendix A.

C.1 From FOCs to best-response quadratics

Laggard: substituting $V_{01} = [(1 - P_L)P_F V_{11} - \frac{c}{2}x^2]/S$ into $cx = \mu(1 - \lambda z)(V_{11} - V_{01})$ and using $V_{11} - V_{01} = [P_L V_{11} + \frac{c}{2}x^2]/S$ gives, after clearing S and simplifying,

$$G_F(x; z) = \frac{c\mu}{2}(1 - \lambda z)x^2 + c\lambda zx - \mu\lambda z(1 - \lambda z)V_{11} = 0.$$

Leader: substituting $V_{10} = [P_L W + (1 - P_L)P_F V_{11} - \frac{c}{2}z^2]/S$ into $cz = \lambda[W - (1 - \mu x)V_{10} - \mu x V_{11}]$; the W -terms simplify using $S - (1 - \mu x)\lambda z = \mu x$ and the V_{11} -terms using $(1 - \mu x)(1 - \lambda z)\mu x + \mu x S = \mu x$, yielding

$$G_L(z; x) = \frac{c\lambda}{2}(1 - \mu x)z^2 + c\mu xz - \lambda\mu x(W - V_{11}) = 0.$$

Root selection: each quadratic has root product $C/A < 0$ (laggard: $-2\lambda z V_{11}/c$; leader: $-2\mu x(W - V_{11})/[c(1 - \mu x)]$, negative given $W > V_{11}$, proven in C.2), so exactly one positive root exists — the additive root. Each firm's best response is therefore the additive root of its quadratic (verified symbolically).

C.2 Value-ordering lemma

Lemma (3, Value ordering, restated). *Under (A1)–(A2), in the MPE: $0 \leq V_{01} \leq V_{11} < W/2$ and $V_{11} \leq V_{10} \leq W$.*

1. $V_{01} \geq 0$: the deviation $x = 0$ yields value 0; optimality of x^* implies $V_{01} \geq 0$.
2. $V_{01} \leq V_{11}$: $S - (1 - P_L)P_F = P_L \geq 0$, so $V_{01} = [(1 - P_L)P_F V_{11} - \frac{c}{2}x^2]/S \leq [SV_{11}]/S =$

V_{11} .

3. $V_{11} < W/2$: evaluate the (1, 1) quadratic at $\tilde{x} = \lambda W(1 - \beta_{11})/c$: $F_{11}(\tilde{x}) = -\lambda W(1 - \beta_{11}) < 0$ (exact identity), so $x_{11}^* < \tilde{x}$, i.e. $cx_{11}^* < \lambda W(1 - \beta_{11})$, and

$$V_{11} = \frac{cx_{11}^*}{2\lambda(1 + \beta_{11})} + \frac{\beta_{11}W}{1 + \beta_{11}} < \frac{W(1 - \beta_{11})}{2(1 + \beta_{11})} + \frac{\beta_{11}W}{1 + \beta_{11}} = \frac{W}{2}.$$

4. $V_{10} \geq V_{11}$ (*mimicry*): let the leader play the stationary deviation $z = x_{11}^*$ against the laggard's MPE strategy x^* . Its value $\tilde{V}$ satisfies $\tilde{V} - V_{11} = x_{11}^* [\lambda(W - V_{11}) - \frac{c}{2}x_{11}^*] / \tilde{S} > 0$ because the symmetric (1, 1) FOC gives $cx_{11}^* = \lambda W - MV_{11} - M^2x_{11}^*(\frac{W}{2} - V_{11}) < \lambda W - \lambda V_{11}$ (using $V_{11} < W/2$, $M \geq \lambda$, $V_{11} \geq 0$). Optimality: $V_{10} \geq \tilde{V} \geq V_{11}$.

5. $V_{10} \leq W$: $V_{10} \leq [P_L W + (1 - P_L)P_F W]/S = W$, using $V_{11} \leq W$. $\square$

C.3 Proof of leader dominance (Proposition 3): $z^* > x^*$ for $\mu \leq \lambda$

The single-crossing argument used in the $\mu = \lambda$ case does not extend to $\mu < \lambda$: pointwise best-response dominance $Z(y) > X(y)$ fails near $y = 0$ once $\mu < \lambda$, because the small- y expansions give $Z(y) \sim \sqrt{2\mu y(W - V_{11})/c}$ and $X(y) \sim \sqrt{2\lambda y V_{11}/c}$, so pointwise dominance there would require $V_{11} < \mu W/(\lambda + \mu)$, strictly stronger than the available bound $V_{11} < W/2$. The following proof works at the equilibrium instead.

Write $B_L \equiv \frac{\lambda}{2}(1 - \mu x)z + \mu x$ and $B_R \equiv \frac{\mu}{2}(1 - \lambda z)x + \lambda z$, both strictly positive for interior efforts, and set

$$L(z, x) \equiv z^2(1 - \lambda z)B_L > 0, \quad R(z, x) \equiv x^2B_R > 0.$$

Step 1 (equilibrium ratio). The leader quadratic $G_L = 0$ factors as $cz B_L = \lambda \mu x(W - V_{11})$, so $L = z^2(1 - \lambda z)B_L = \kappa(W - V_{11})$ with $\kappa \equiv \lambda \mu x z(1 - \lambda z)/c > 0$. The laggard quadratic $G_F = 0$ factors as $cx B_R = \mu \lambda z(1 - \lambda z)V_{11}$, so $R = x^2 B_R = \kappa V_{11}$. Hence at any interior

equilibrium

$$\frac{R}{L} = \frac{V_{11}}{W - V_{11}} < 1,$$

the inequality by $V_{11} < W/2 \Rightarrow V_{11} < W - V_{11}$ (Lemma 3). Thus $R < L$ at equilibrium.

Step 2 (algebraic inequality). For all (z, x) with $0 < \lambda z < 1$, $0 < \mu x < 1$, $x \geq z$, and $0 < \mu \leq \lambda$: $\Phi \equiv R - L \geq 0$. The quantity Φ is affine in μ ($\partial^2 \Phi / \partial \mu^2 \equiv 0$), so on $\mu \in [0, \lambda]$ it is the convex combination $\Phi(\mu) = (1 - \frac{\mu}{\lambda})\Phi(0) + \frac{\mu}{\lambda}\Phi(\lambda)$; it suffices that both endpoints are nonnegative for $x \geq z$:

- $\Phi|_{\mu=0} = \lambda z(x^2 - \frac{1}{2}(1 - \lambda z)z^2) \geq \lambda z(z^2 - \frac{1}{2}z^2) = \frac{1}{2}\lambda z^3 > 0$;
- $\Phi|_{\mu=\lambda} = \lambda \psi(z, x)$ with $\psi(z, x) \equiv x^2[\frac{1}{2}(1 - \lambda z)x + z] - z^2(1 - \lambda z)[\frac{1}{2}(1 - \lambda x)z + x]$, where ψ is strictly increasing in x on $x \geq z$ (since $\partial^2 \psi / \partial x^2 = 3(1 - \lambda z)x + 2z > 0$ and $\partial \psi / \partial x|_{x=z} = \frac{1}{2}z^2(5 - \lambda^2 z^2) > 0$) and $\psi(z, z) = \frac{1}{2}\lambda z^4(3 - \lambda z) > 0$; hence $\psi(z, x) > 0$.

So $\Phi \geq 0$, i.e. $R \geq L$, whenever $x \geq z$. The hypothesis $\mu \leq \lambda$ enters *only here*, keeping μ inside $[0, \lambda]$; for $\mu > \lambda$ the affine extension of Φ can turn negative, which is the analytic source of the one-sided necessity.

Step 3 (contradiction). Suppose $x^* \geq z^*$ at the interior equilibrium. Step 2 gives $R(z^*, x^*) \geq L(z^*, x^*)$, contradicting $R < L$ from Step 1. Hence $z^* > x^*$. ■

All identities in this subsection (the factorizations of G_L, G_F ; $\partial^2 \Phi / \partial \mu^2 = 0$; the two endpoint expressions; $\psi(z, z)$ and the derivatives of ψ) were verified symbolically in `prop2_extension_symbolic` and `prop2_dag_identity.py`; the inequality $R \geq L$ for $x \geq z$, $\mu \leq \lambda$ was checked over 3×10^6 random draws (zero violations) and $z^* > x^*$ over 60,000 equilibrium draws with $\mu \leq \lambda$ (zero failures).

Remark. *The hypothesis is one-sided. Leader dominance never fails for $\mu \leq \lambda$; among $\mu \neq \lambda$ draws it fails in $\approx 22\%$, all with $\mu > \lambda$. So $\mu \leq \lambda$ is sufficient, and only $\mu > \lambda$ can overturn the ordering.*

C.4 Non-monotone laggard response

Eliminating V_{11} from $G_{F,z}$ via $G_F = 0$ yields the exact identity

$$G_{F,z}|_{BR} = -\frac{cx N}{2z(1-\lambda z)}, \quad N \equiv \mu x(1-\lambda z)^2 - 2\lambda^2 z^2,$$

so $\text{sign}(X'(z)) = \text{sign}(N)$: *provocation* for small z (the fall in V_{01} dominates), *discouragement* for large z . Consequently X is not globally monotone and no monotone fixed-point argument is available; the transmission results in C.5 instead proceed through the Implicit Function Theorem, which requires only the regularity established there.

C.5 Regularity, uniqueness, and transmission

Let J denote the Jacobian of (G_L, G_F) with respect to (z, x) at an interior solution of $G_L = G_F = 0$.

Lemma 7 (Regularity). *Under (A1), at any interior fixed point $(z^*, x^*) \in (0, 1)^2$,*

$$\det J = c^2 \left[(\lambda(1-\mu x)z + \mu x)(\mu(1-\lambda z)x + \lambda z) - \frac{\lambda\mu x z(1-\lambda z)}{4} + \frac{\lambda^3 z^3}{2(1-\lambda z)} \right] > 0.$$

Proof. $G_{L,z} = c\lambda(1-\mu x)z + c\mu x$ and $G_{F,x} = c\mu(1-\lambda z)x + c\lambda z$ contain neither W nor V_{11} , and the cross-partial at the fixed point are the exact identities $G_{L,x}|_{BR} = -c\lambda z^2/(2x)$, which follows from eliminating $W - V_{11}$ from $\partial G_L/\partial x$ using $G_L = 0$ as in C.1, and $G_{F,z}|_{BR} = -cxN/(2z(1-\lambda z))$ from C.4. Substituting into $\det J = G_{L,z}G_{F,x} - G_{L,x}G_{F,z}$ gives the display. Expanding the product and collecting $\lambda\mu x z - \lambda\mu x z(1-\lambda z)/4 = \lambda\mu x z(3+\lambda z)/4$,

$$\frac{\det J}{c^2} = \lambda\mu x z(1-\mu x)(1-\lambda z) + \lambda^2 z^2(1-\mu x) + \mu^2 x^2(1-\lambda z) + \frac{\lambda\mu x z(3+\lambda z)}{4} + \frac{\lambda^3 z^3}{2(1-\lambda z)},$$

and under (A1) every term is strictly positive at an interior fixed point. $\square$

Remark. *Since $Z'(x) = -G_{L,x}/G_{L,z}$ and $X'(z) = -G_{F,z}/G_{F,x}$ with $G_{L,z}, G_{F,x} > 0$, Lemma 7*

gives $Z'X' - 1 = -\det J/(G_{L,z}G_{F,x}) < 0$ at every fixed point: the composite map $z \mapsto Z(X(z))$ crosses the 45° line strictly from above wherever it crosses at all. A continuous map cannot do so twice, so the interior fixed point is unique given existence. This does not establish convergence (stability) of best-response iteration, which remains a numerical observation (Section 5).

Remark (Why existence is left numerical). A fixed-point argument on (G_L, G_F) does not settle existence, because the origin solves both quadratics. It is not an equilibrium — S vanishes there, so the substitution of C.1 is undefined — but it is a fixed point of the substituted system, so Brouwer returns it and nothing else. An interior fixed point needs an index argument at the origin, which I do not attempt.

Proposition 5 (Laggard transmission). *Under (A1)–(A2), at the interior equilibrium, if $x^* < \sqrt{2}z^*$ then $\frac{dx^*}{dV_{11}} > 0$.*

Proof. By Lemma 7 the Implicit Function Theorem applies, and $dx^*/dV_{11} = (G_{L,V}G_{F,z} - G_{L,z}G_{F,V})/\det J$ with $G_{L,V} = \lambda\mu x$ and $G_{F,V} = -\mu\lambda z(1 - \lambda z)$. Substituting the exact identities and simplifying, the numerator equals

$$\frac{c\lambda\mu}{2z(1 - \lambda z)} \left[2\lambda^2 x^2 z^2 + 2\lambda(1 - \mu x)z^3(1 - \lambda z)^2 + \mu x(1 - \lambda z)^2(2z^2 - x^2) \right].$$

The first two bracketed terms are strictly positive under (A1); the third is nonnegative exactly when $x \leq \sqrt{2}z$. Hence the numerator is strictly positive whenever $x^* < \sqrt{2}z^*$, and the sign follows from $\det J > 0$. $\square$

Corollary 1 (Transmission of β_{11} for $\mu \leq \lambda$). *Assume $\mu \leq \lambda$. Then $\frac{dx^*}{d\beta_{11}} > 0$.*

Proof. Spillovers are inactive in the asymmetric state, so β_{11} enters (G_L, G_F) only through V_{11} , and $dx^*/d\beta_{11} = (dx^*/dV_{11}) \cdot (\partial V_{11}/\partial \beta_{11})$. For $\mu \leq \lambda$, Proposition 3 gives $z^* > x^*$, hence $x^* < z^* < \sqrt{2}z^*$ and Proposition 5 applies; $\partial V_{11}/\partial \beta_{11} > 0$ is Proposition 2. $\square$

Lemma 8 (Leader sign). *At the interior equilibrium,*

$$\text{sign}\left(\frac{dz^*}{d\beta_{11}}\right) = \text{sign}\left(\frac{dz^*}{dV_{11}}\right) = \text{sign}\left(\frac{1}{2}\lambda z^3(1 - \lambda z) - \mu(1 - \lambda z)x^3 - \lambda z x^2\right).$$

Proof. The Cramer numerator $-G_{L,V}G_{F,x} + G_{L,x}G_{F,V}$ simplifies, via the same identities, to $(c\lambda\mu/x)\left[\frac{1}{2}\lambda z^3(1 - \lambda z) - \mu(1 - \lambda z)x^3 - \lambda z x^2\right]$; the claim follows from $\det J > 0$ and $\partial V_{11}/\partial\beta_{11} > 0$. $\square$

All displayed identities in C.4–C.5 were verified to symbolic zero (`verify_step8_transmission.py`) and cross-checked against finite differences at 20,000 solved equilibria.

C.6 Total-effect comparative statics

For $\theta \in \{c, \lambda\}$: $\frac{d(z^*, x^*)}{d\theta} = -J^{-1}\left[\frac{\partial G}{\partial\theta} + \frac{\partial G}{\partial V_{11}}\frac{\partial V_{11}}{\partial\theta}\right]$ (well-defined by Lemma 7), with $\partial V_{11}/\partial c > 0$ and $\partial V_{11}/\partial\lambda < 0$ from Appendix B. Scan results (seed 11, $n = 8,000$ subsample): $dz^*/dc < 0$ and $dx^*/dc < 0$ in all 8,000 draws, with direct and total signs agreeing everywhere. $dz^*/d\lambda > 0$ in 7,999 draws; the one exception is a direct-versus-total sign reversal. Of the 163 draws with $dx^*/d\lambda < 0$, 157 are reversals where the indirect V_{11} channel overturns a positive direct effect (1.96% of the subsample) and the remaining six have a negative direct effect as well. A second scan (seed 202; an $n = 12,000$ subsample of the 60,000-draw confirmation battery) puts the reversal rate at 1.76%, and a third (seed 707, same design) at 1.93%.

D Initial Research State: Derivation

Spillovers active: $P_i = \mu(x_i + \beta_{00}x_j)$, $P_j = \mu(x_j + \beta_{00}x_i)$; $m \equiv \mu(1 + \beta_{00})$.

D.1 Bellman, FOC, and the equilibrium quadratic

Bellman:

$$V_{00} = -\frac{c}{2}x_i^2 + P_i(1 - P_j)V_{10} + P_j(1 - P_i)V_{01} + P_iP_jV_{11} + (1 - P_i)(1 - P_j)V_{00},$$

isolated as $V_{00} = [P_i(1 - P_j)V_{10} + P_j(1 - P_i)V_{01} + P_iP_jV_{11} - \frac{c}{2}x_i^2]/(P_i + P_j - P_iP_j)$. The FOC (continuation value fixed) is

$$cx_i = \mu[(1 - P_j - \beta_{00}P_i)V_{10} + (\beta_{00}(1 - P_i) - P_j)V_{01} + (P_j + \beta_{00}P_i)V_{11} - ((1 - P_j) + \beta_{00}(1 - P_i))V_{00}].$$

Under symmetry ($x_i = x_j = x$, $P = mx$) the FOC becomes

$$cx = \mu(1 - P)(V_{10} + \beta_{00}V_{01}) - \mu P(\beta_{00}V_{10} + V_{01}) + mPV_{11} - m(1 - P)V_{00},$$

and the Bellman-form value is $V_{00} = [P(1 - P)(V_{10} + V_{01}) + P^2V_{11} - \frac{c}{2}x^2]/[P(2 - P)]$.

Equating the FOC-implied and Bellman-implied V_{00} and clearing denominators yields (with $G \equiv V_{10} + \beta_{00}V_{01} - (1 + \beta_{00})V_{11}$, $D \equiv V_{10} - V_{01}$)

$$\boxed{cmx^2 - [3c + 2\mu mG]x + 2\mu(1 - \beta_{00})D = 0,}$$

with x_{00}^* the subtracted root and equilibrium value

$$V_{00} = \frac{\mu(V_{10} + \beta_{00}V_{01}) - mP(V_{10} + V_{01} - V_{11}) - cx_{00}^*}{m(1 - mx_{00}^*)}.$$

D.2 Master quadratic lemma

Lemma (4, Master quadratic, restated). *Substituting $(V_{10}, V_{01}, V_{11}) \rightarrow (W, 0, W/2)$, $\mu \rightarrow \lambda$, $\beta_{00} \rightarrow \beta_{11}$ into the boxed quadratic reproduces the (1, 1) equilibrium quadratic exactly ($G \rightarrow W(1 - \beta_{11})/2$ gives the middle coefficient $3c + \lambda^2W(1 - \beta_{11}^2)$ and constant $2\lambda W(1 - \beta_{11})$);*

the same substitution maps the V_{00} expression to the (1,1) FOC-form value. One quadratic solves both symmetric states.

D.3 Root selection and regularity

Lemma (5, Root selection, (0,0), restated). *Under (A1), (A2') $\mu W \leq c$, and the value ordering of Lemma 3:*

$$mF_{00}(1/m) = -2c - 2\mu m[\beta_{00}(V_{10} - V_{11}) - (V_{11} - V_{01})] \leq -2c + 2\mu mW \leq -2c + 2mc < 0,$$

so the roots straddle $1/m$: the additive root implies $P > 1$ (infeasible); the subtracted root is the unique valid candidate. Both roots are strictly positive: the product is $2\mu(1 - \beta_{00})D/(cm) > 0$ (by $V_{10} > V_{01}$) and the middle coefficient satisfies $2\mu m|G| \leq 2\mu mW \leq 2mc < 3c$, so $3c + 2\mu mG > 0$.

Real roots. $\Delta_0 = (3c + 2\mu mG)^2 - 8cm\mu(1 - \beta_{00})D > 0$ is proven analytically for $m < \frac{1}{2}$ (via $(3 - 2m)^2 \geq 8m$) and for $\beta_{00} \geq \frac{4}{5}$ (via $(2 - \beta_{00})^2 \geq 4(1 - \beta_{00}^2)$); with the value functions evaluated at the full equilibrium, it holds in all 60,000 scanned draws ($\min \Delta_0 \approx 2 \times 10^{-2}$).

Interiority $x_{00}^* \in (0, 1)$, $mx_{00}^* < 1$: zero violations in the same scan. Where the analytical proof does not apply, the claim rests on this numerical check.

D.4 Small- μ limit

Series expansion of the subtracted root: $x_{00}^* = \frac{2\mu(1 - \beta_{00})(V_{10} - V_{01})}{3c} + O(\mu^2)$ — the simple proportional-to-the-value-gap formula from the mutually-exclusive-breakthrough model is the rare-breakthrough limit of the present framework. $\square$